

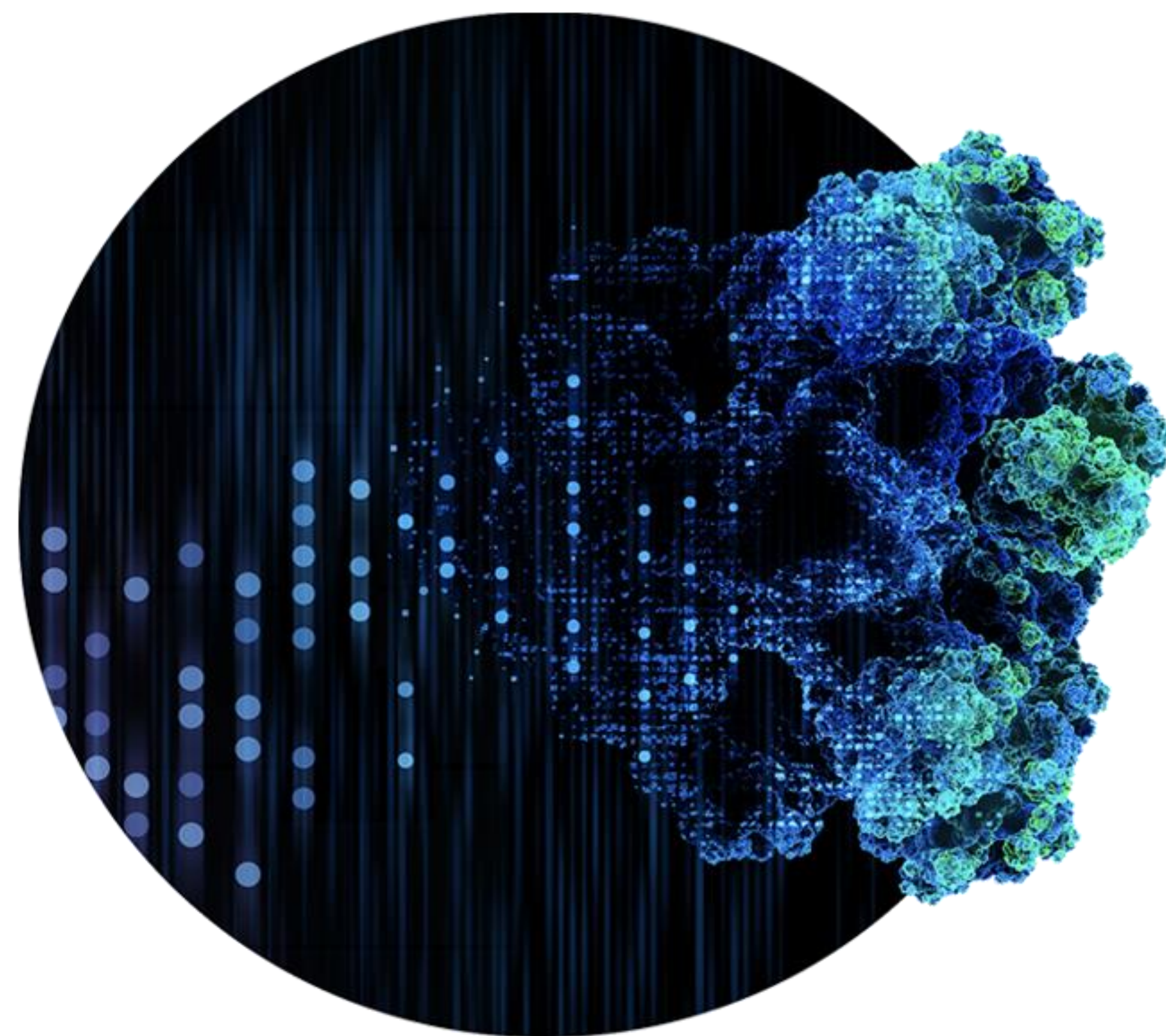
ΔTissue Scientific Progress Update: *Brief for Wellcome BoG Meeting- FY24Q3*

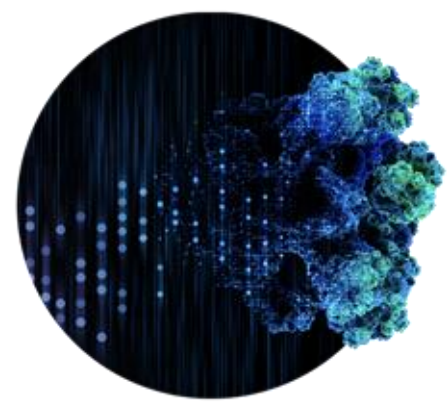
Jason Swedlow
Program Director

Josh Titlow
SETA

5 / 24/ 2024

CONFIDENTIAL

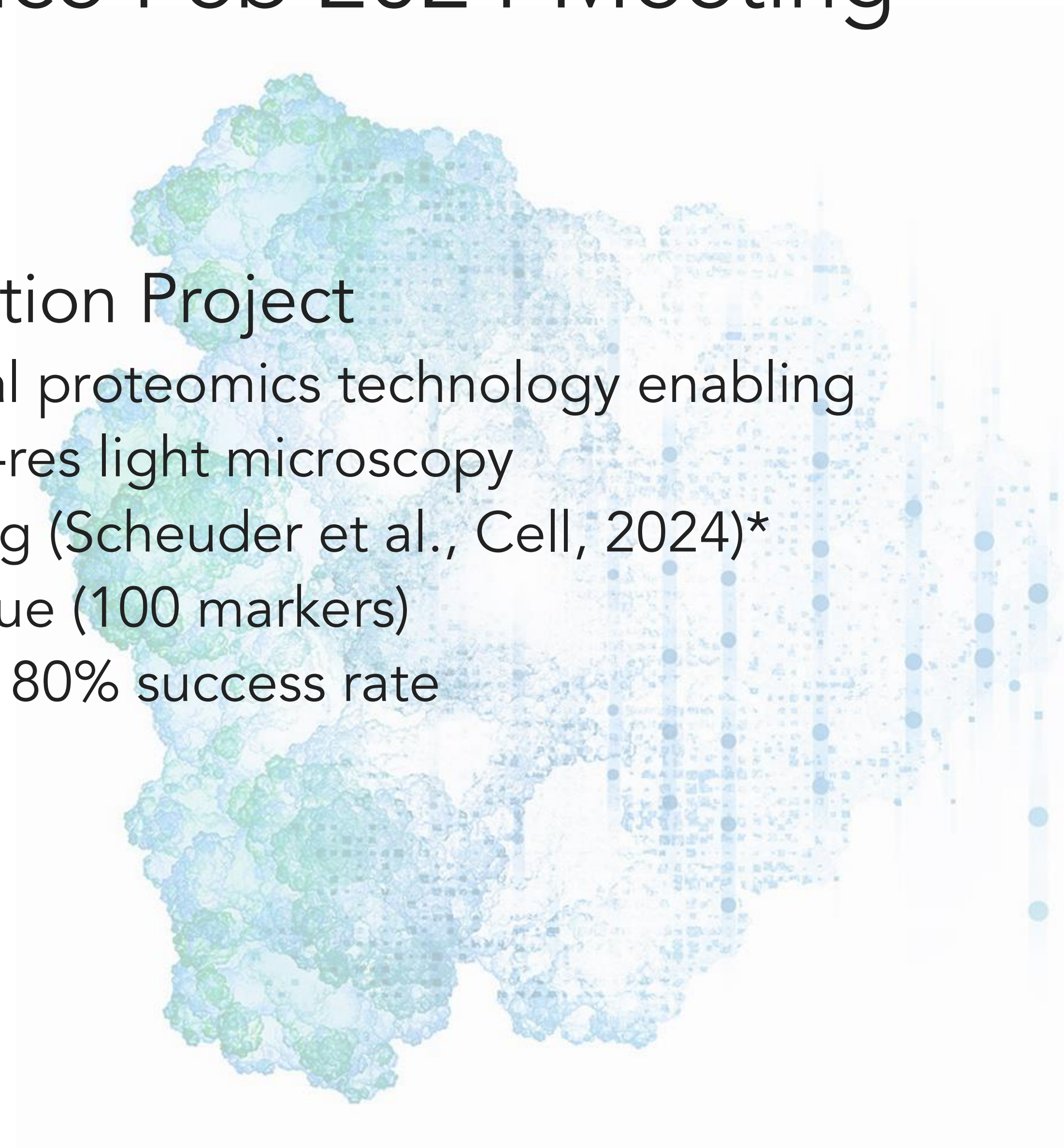




Summary of Progress since Feb 2024 Meeting

Technology Development- Yale Transition Project

- Flash-PAINT is a potentially disruptive spatial proteomics technology enabling theoretically unlimited multiplexing for high-res light microscopy
 - Demonstrated capability of 9-plex imaging (Scheuder et al., Cell, 2024)*
 - Δ Tissue goal is to scale it 10x in FFPE tissue (100 markers)
 - Currently validated over 30 adapters with 80% success rate



FLASH-PAINT:

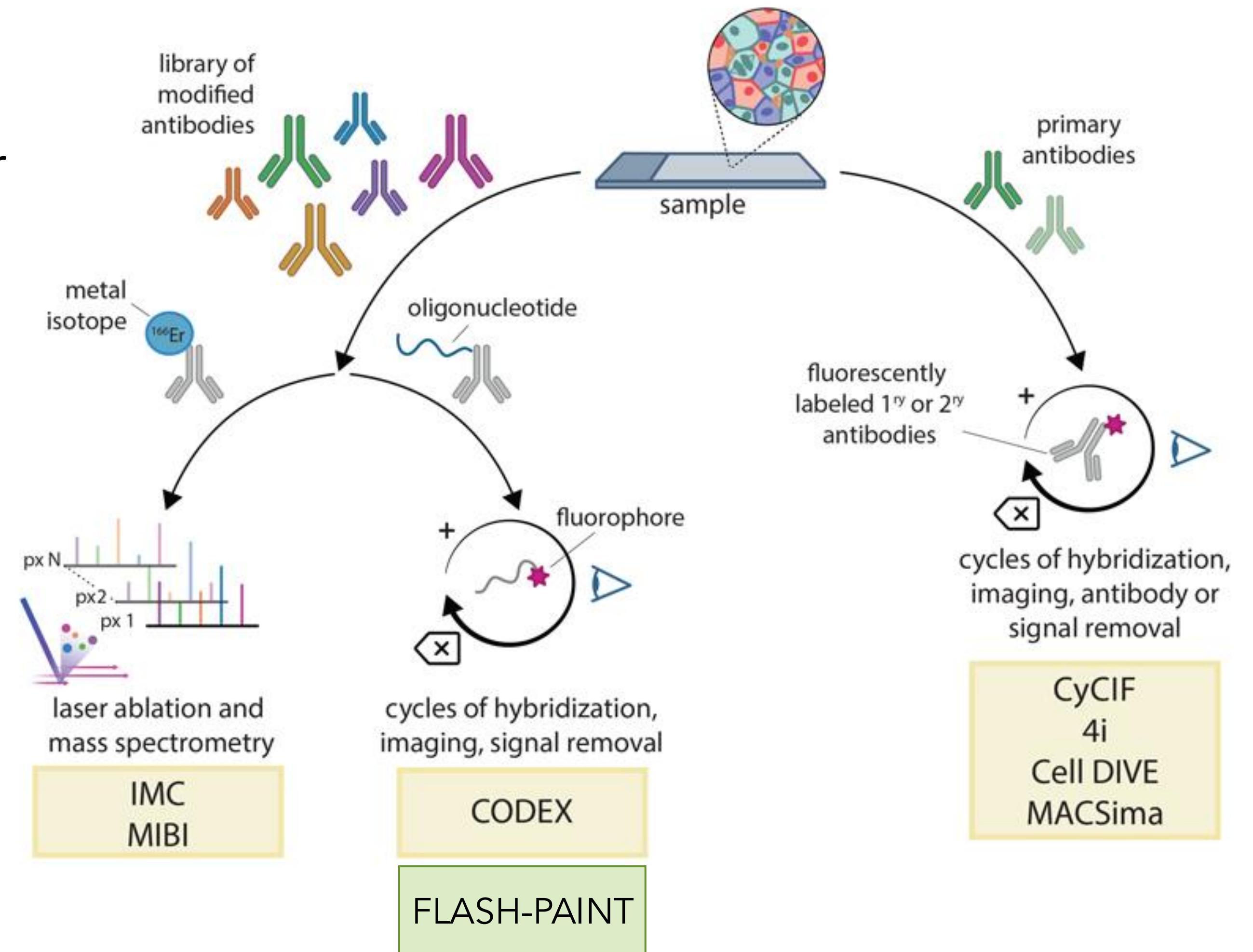
Fast, gentle, and efficient multiplexed imaging

Presented by Sylvi Stoller
Asmita Jha and Sylvi Stoller



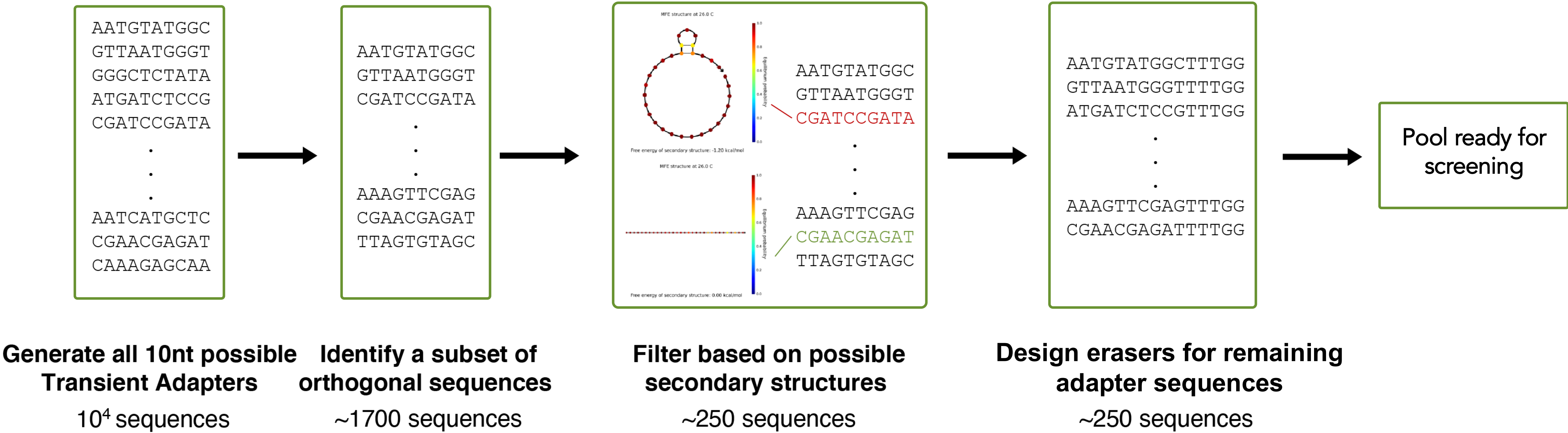
Spatial omics

- Spatial omics - measuring physical location and structure and molecular characteristics at the same time
 - Sequencing-based methods
 - Imaging-based methods
- Challenges:
 - High multiplexing
 - Throughput
 - Bulk data analysis



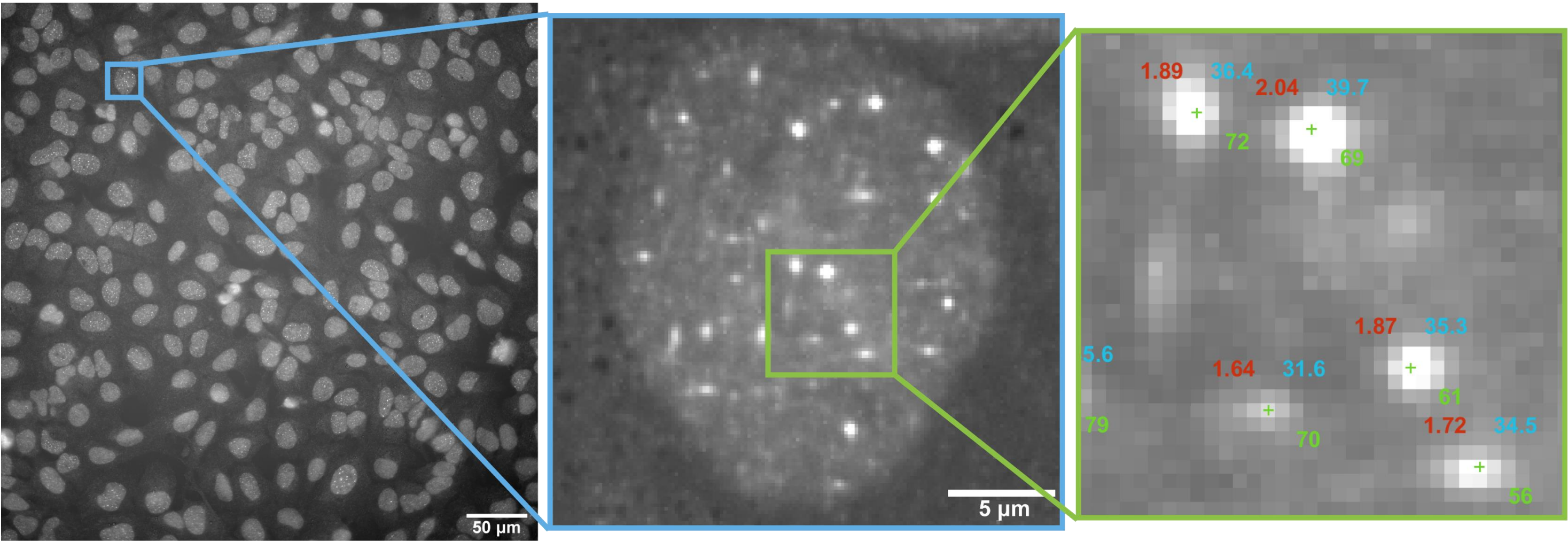
Expanding FLASH-PAINT to 100+ Multiplexing

Generating Transient Adapter Pool

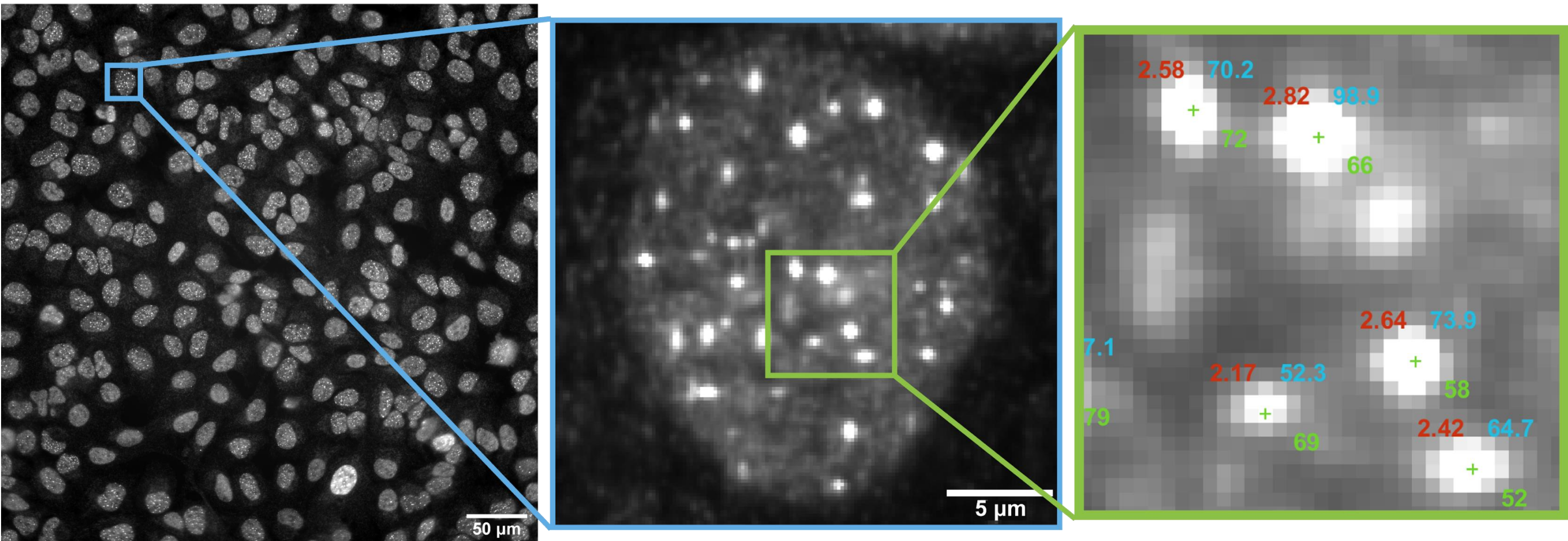


Expanding FLASH-PAINT to 100+ Multiplexing

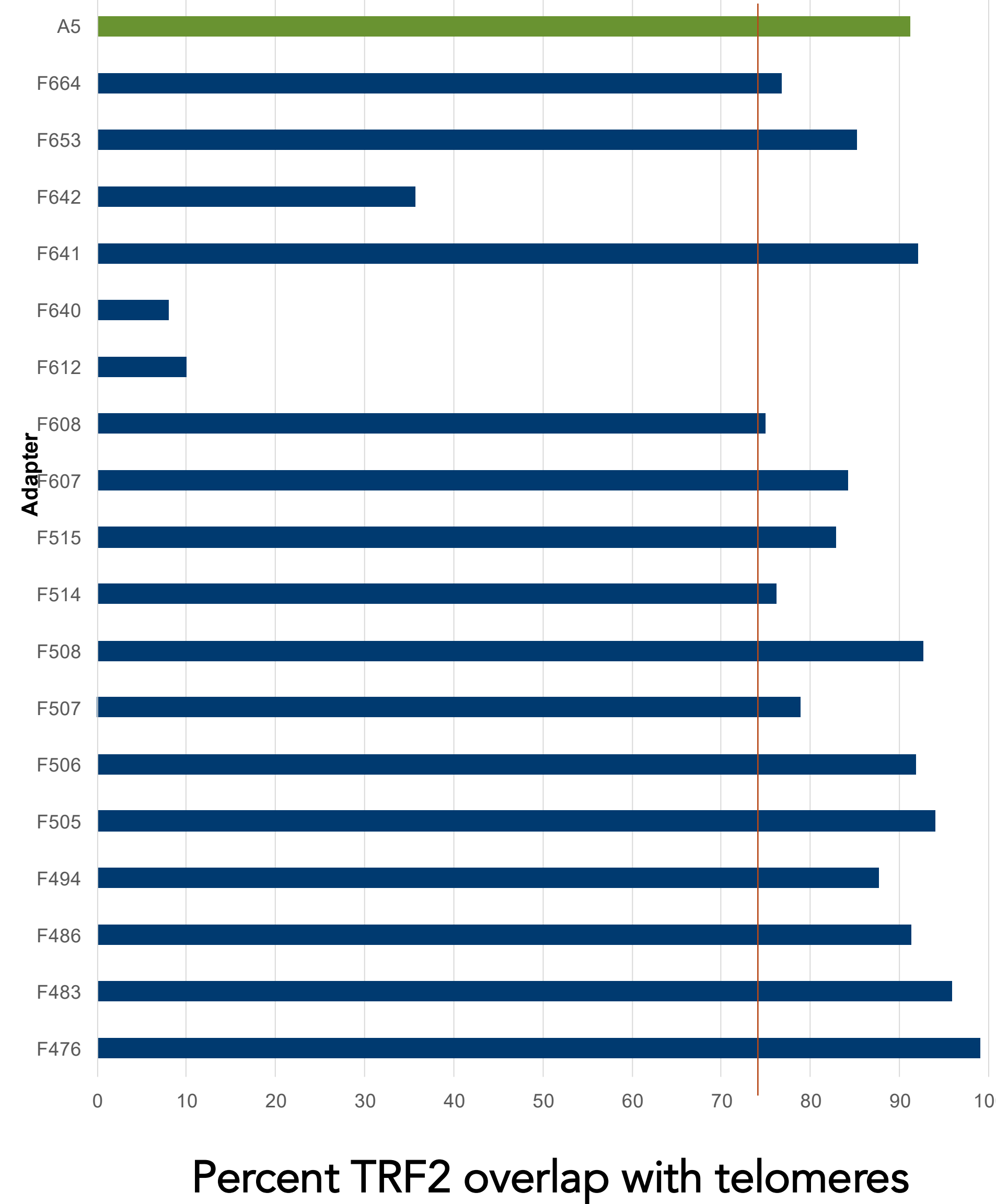
a. Telomere FISH with Transient Adapter

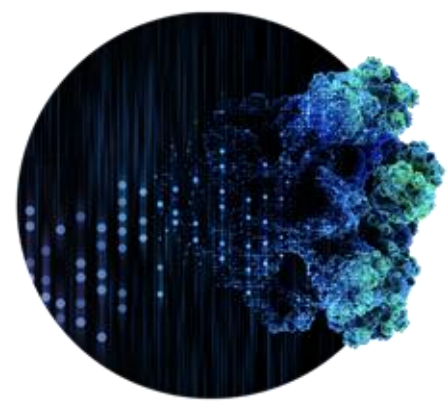


b. TRF2 with Atto647n



c. Transient Adapter Screen Results



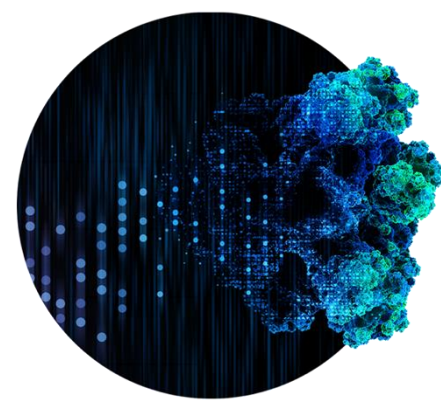


Summary of Progress since Feb 2024 Meeting

TNBC- Tons of multi-modal data rolling in from all three projects

UCSD Project

- Part 1- A potential drug target for the worst tumors
 - A 42-gene spatial transcriptomics panel distinguishes tissue state transitions (atypia to invasive) in tumors from poor and good outcomes
 - High expression of C3 in atypia regions is highly indicative of poor outcomes
 - Therapies that target C3 have already been approved for use in humans
- Part 2- An RNA-based patient stratification that points to specific therapies for 3 subsets of TNBC patients
 - A 261 gene panel stratifies patients into 3 subgroups
 - Differential RNA expression points to specific pathways and potential drug targets specific for each patient subgroup



How do we enable health breakthroughs?

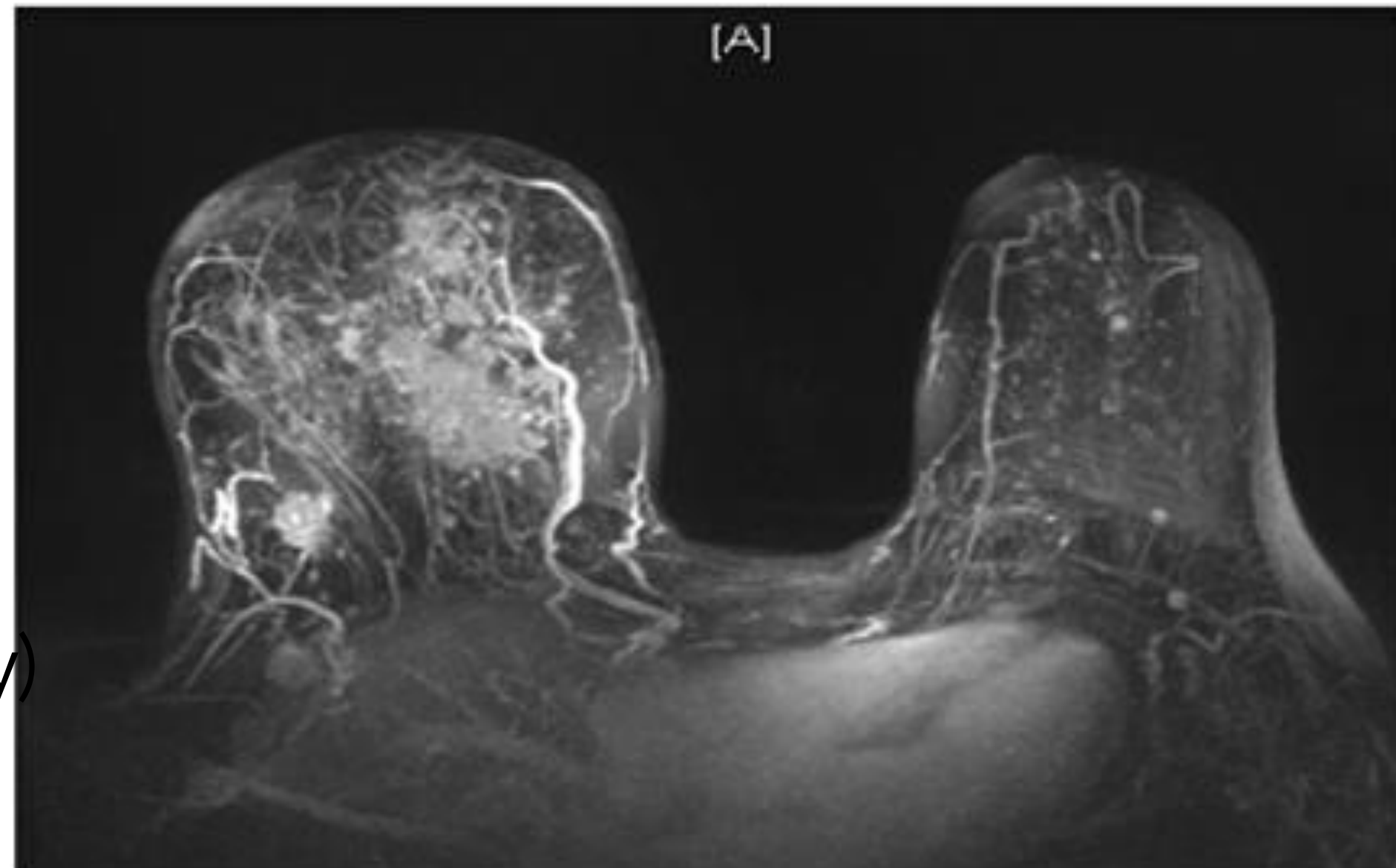
TNBC Data Produced:

40	patient MRE acquisitions
222	patient spatial transcriptomes (>10,000 ROIs)
100	patient metallomes
176	spatial proteomes (35-70 markers)
11	patient-derived organoid lines
3	TNBC Tissue Time Machines
254	patient bulk transcriptomes
51	patient bulk BCR/TCR transcriptomes
129	single cell transcriptome samples (62 patients)
118	single cell genome samples (59 patients)

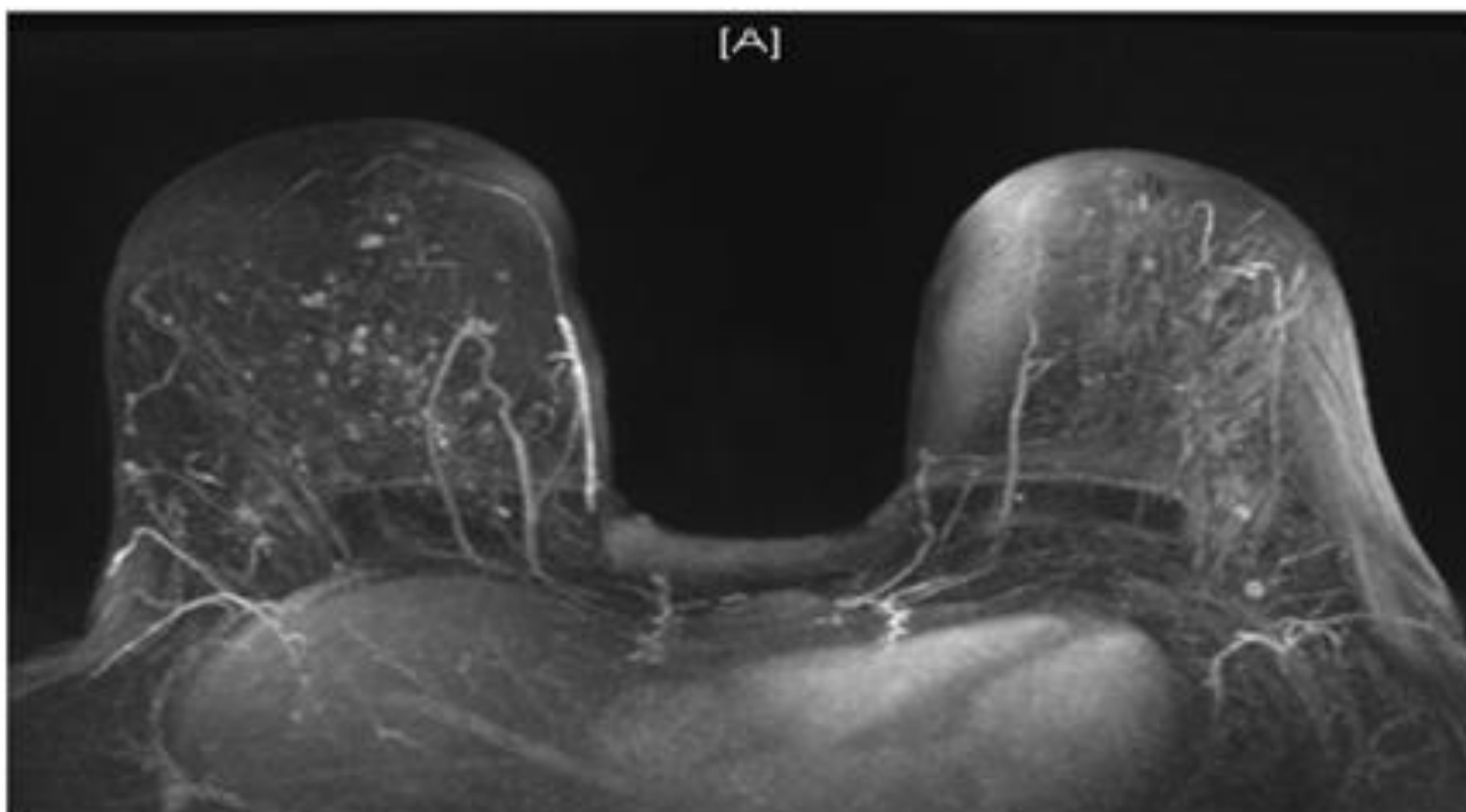
Part 1

Which woman has the aggressive TNBC? Age ~40, AA, locally advanced

Patient #1



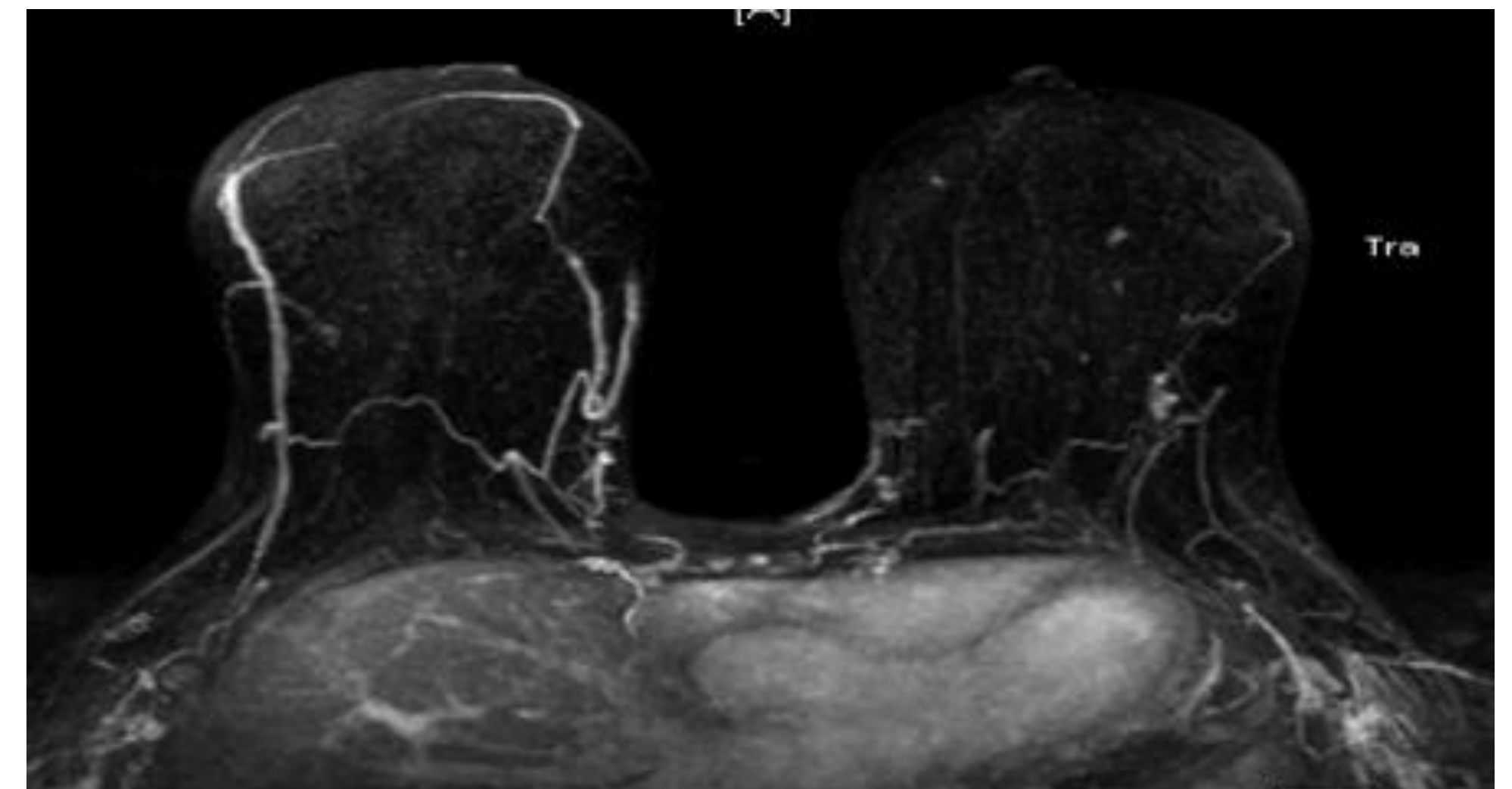
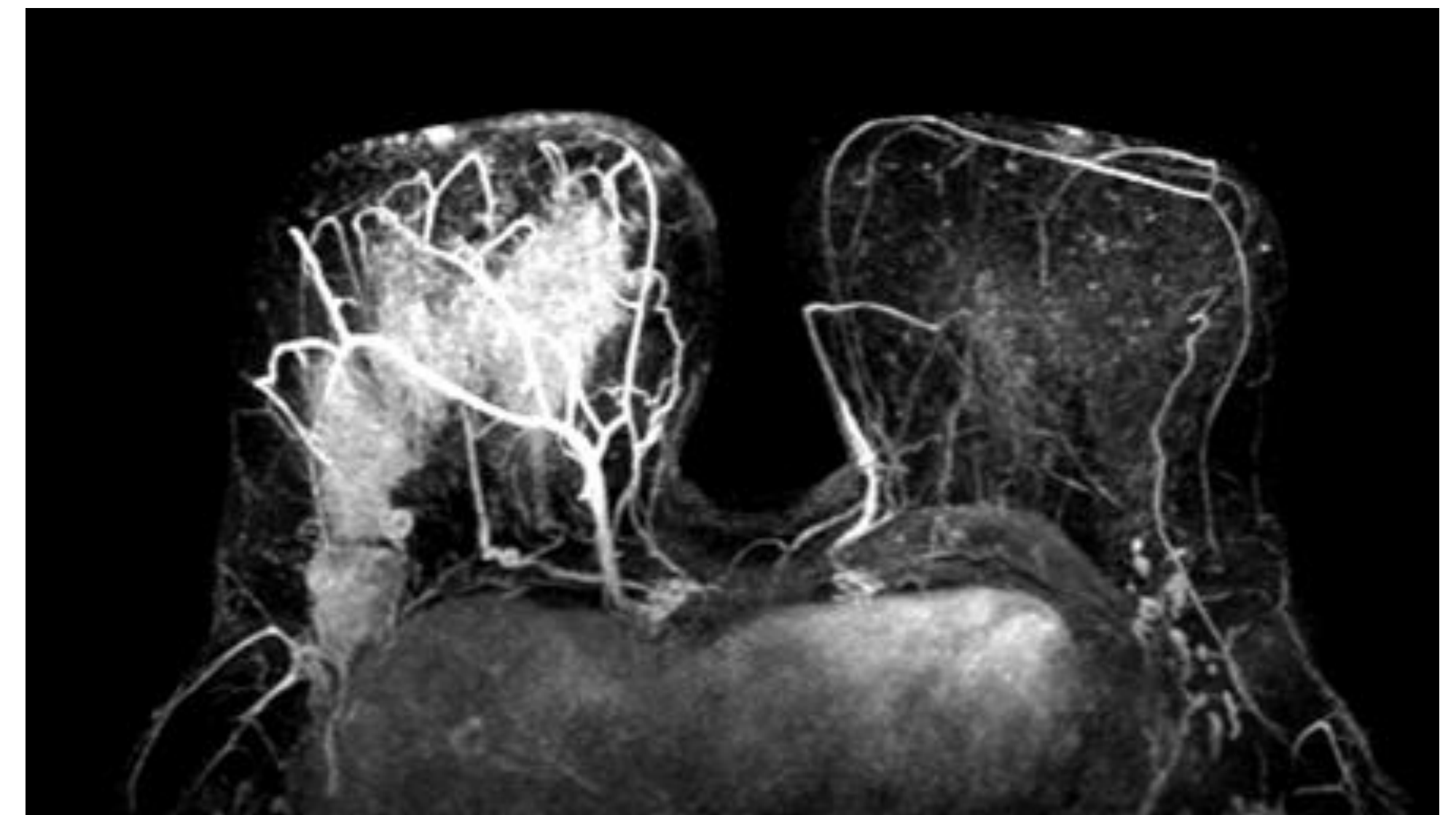
Before NACT
(Neoadjuvant
Chemotherapy)



After NACT
No TNBC

Dead 6 months

Patient #2



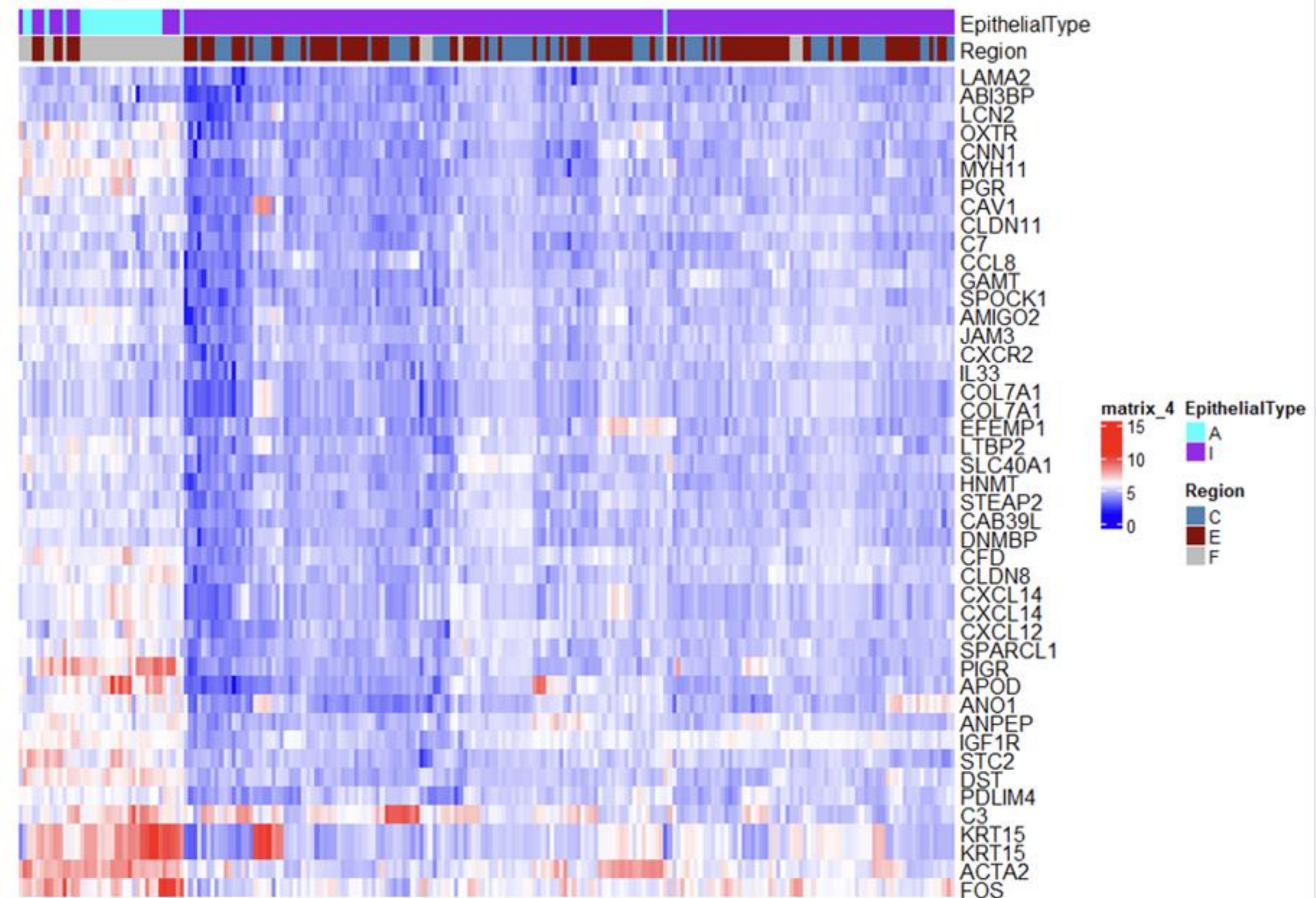
Alive 15 years

| COH/Duke/LSU TNBC Samples

Sample Set	Set Info	Samples Completed	Analysis	Study
LSU TNBC	254 patients, 50/% AA	254	Bulk RNASeq	Predictive Survival Markers
LSU/Duke Good/ Poor Px (prognosis)	64 extreme, residual 59 no residual	64 59	GeoMx	Biomarkers linked to residual disease, aggressive tumours
Duke Pregnancy	Pre-/Post Involution Validation	21 12	GeoMx	Biology of pregnancy/involution

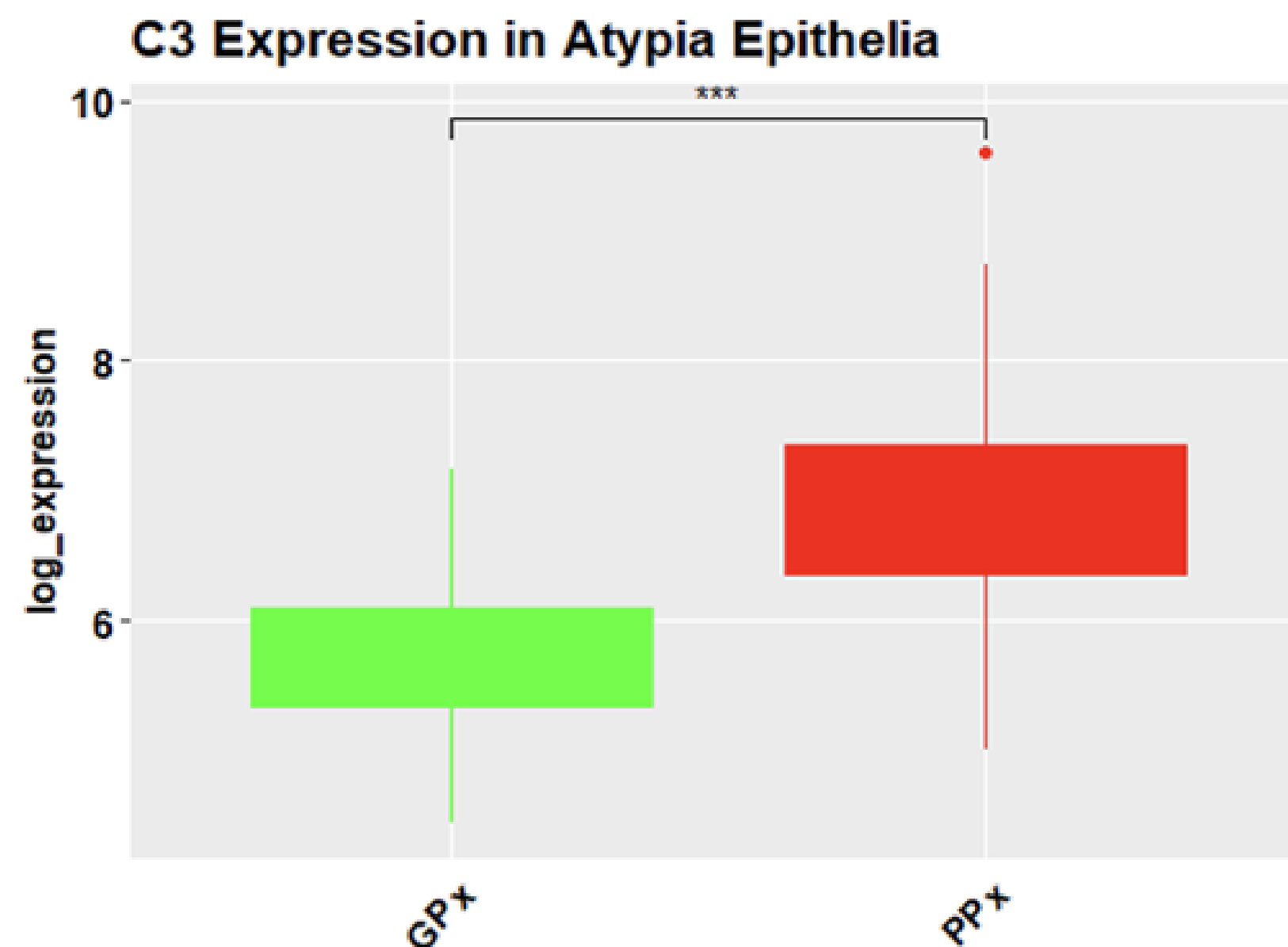
Atypia-specific in PPX

- A 42 gene signature in atypia of PPx that separates the atypia from the invasive state
- Not seen in GPx



Distinct complement activation in Atypia between GPx and PPx

- BulkSignalR on PanCK+ and PanCK- Atypia to discover epithelial-stromal interactions related to C3
- We observe significantly increased C3 expression in the PPX Atypia epithelia



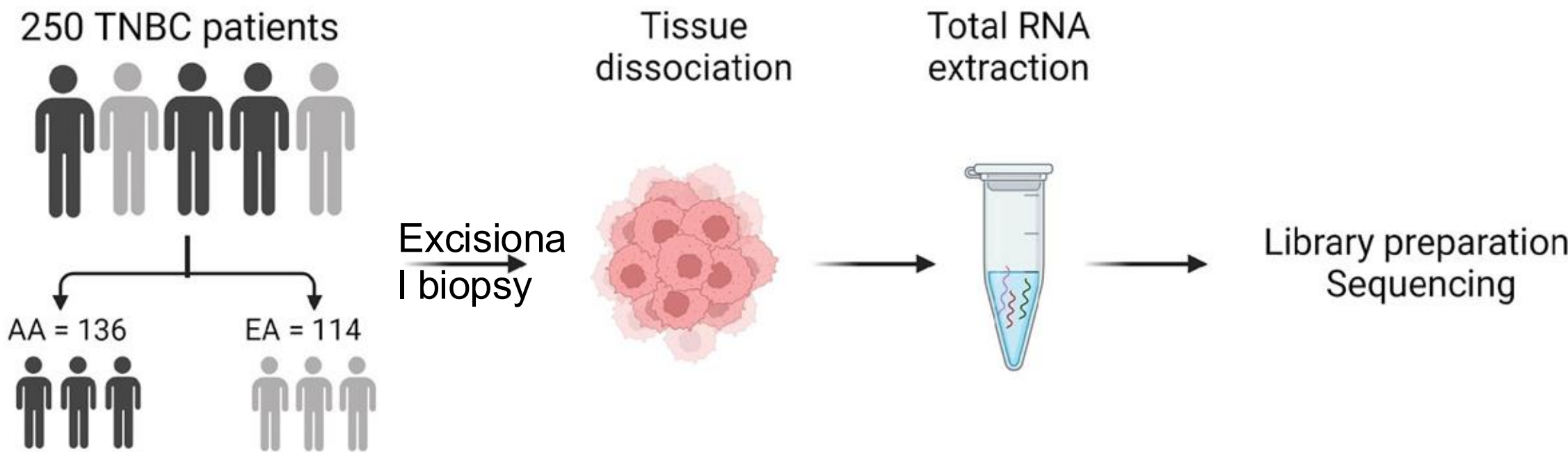
GPx uniquely shows **C3->ITGAX (CR4)** activity. CR4 is commonly expressed in myeloid and NK cells and can induce phagocytosis.

PPx uniquely shows **C3->CD81** activity. CD81 is a tetraspanin associated with cell migration and can be expressed by B cells or on exosomes secreted from CAFs.

Part 2

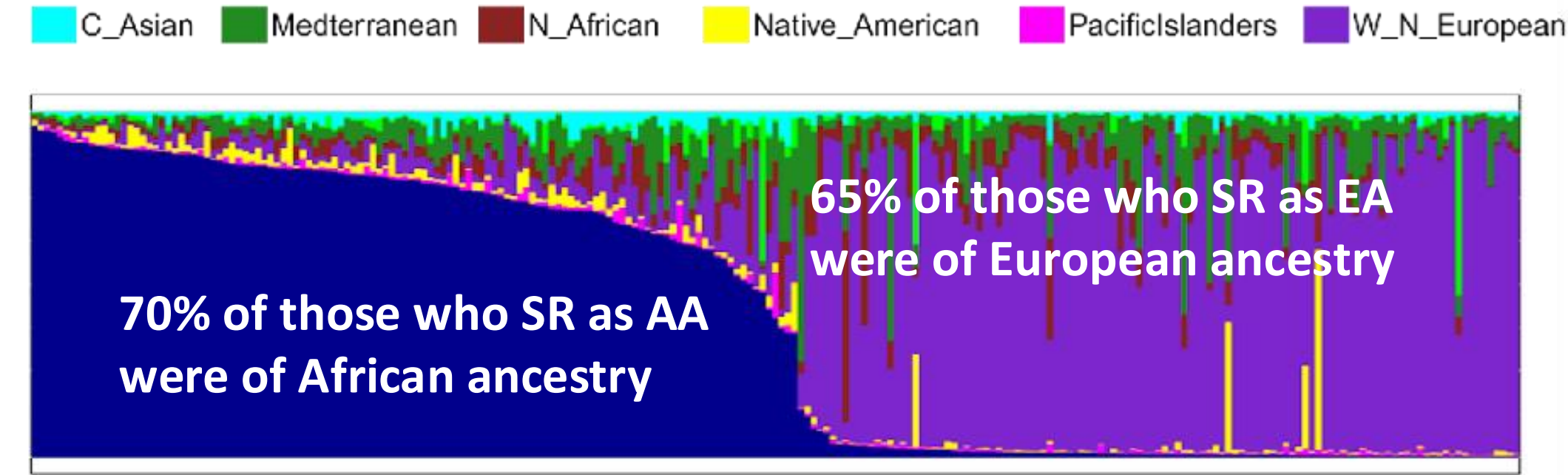
TNBC Survival Study Design

Louisiana State University

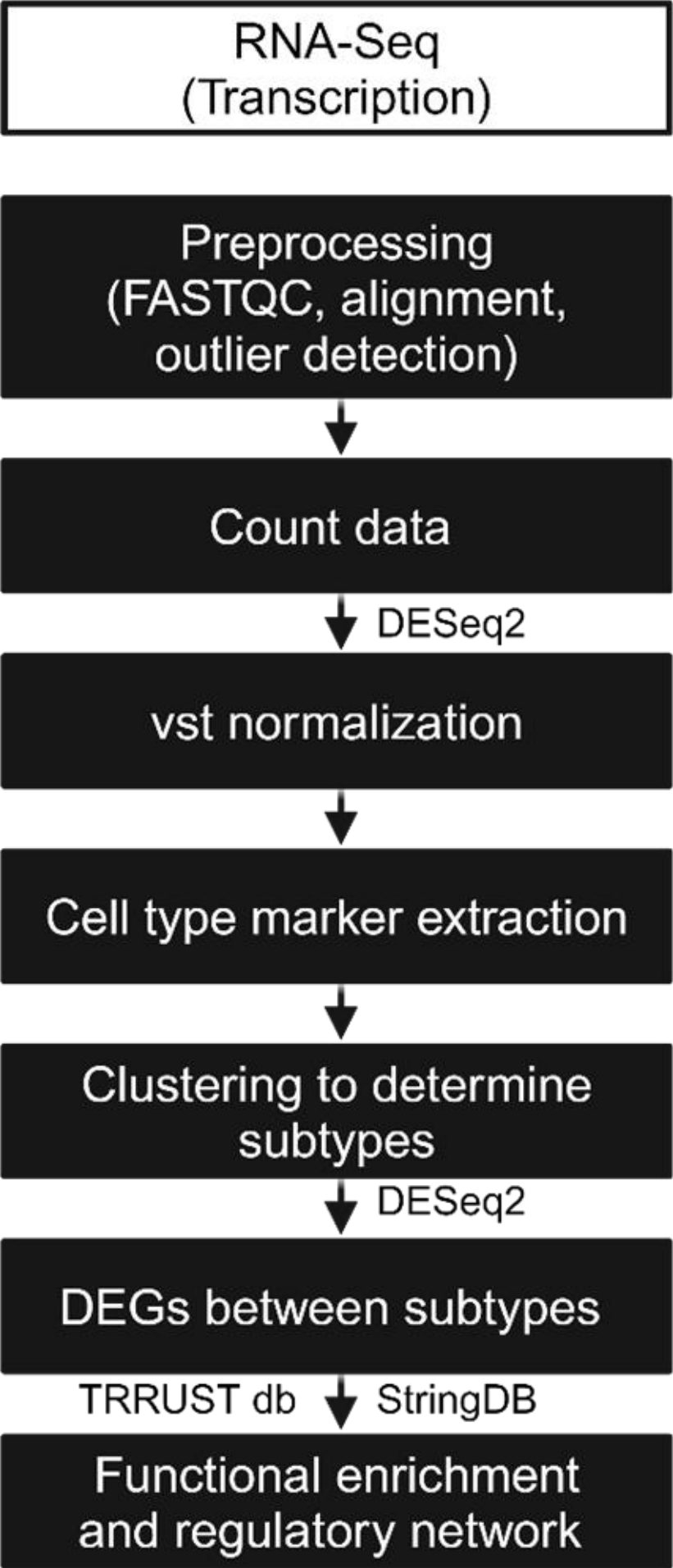


Self-reported-race (SRR)
Stage, Co-morbidity, Survival
Clinical phenotyping

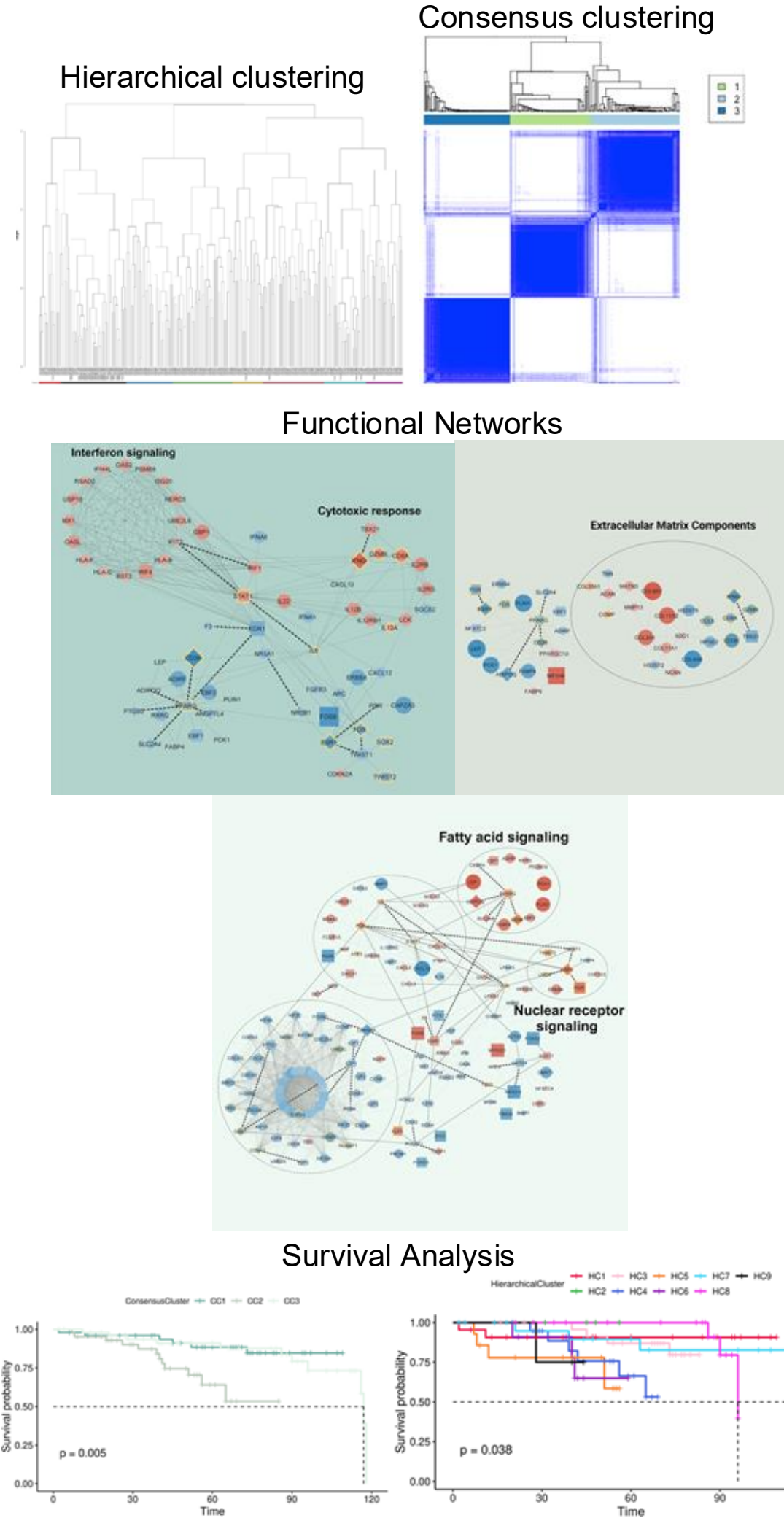
City of Hope
Genetic Admixture (N = 227)



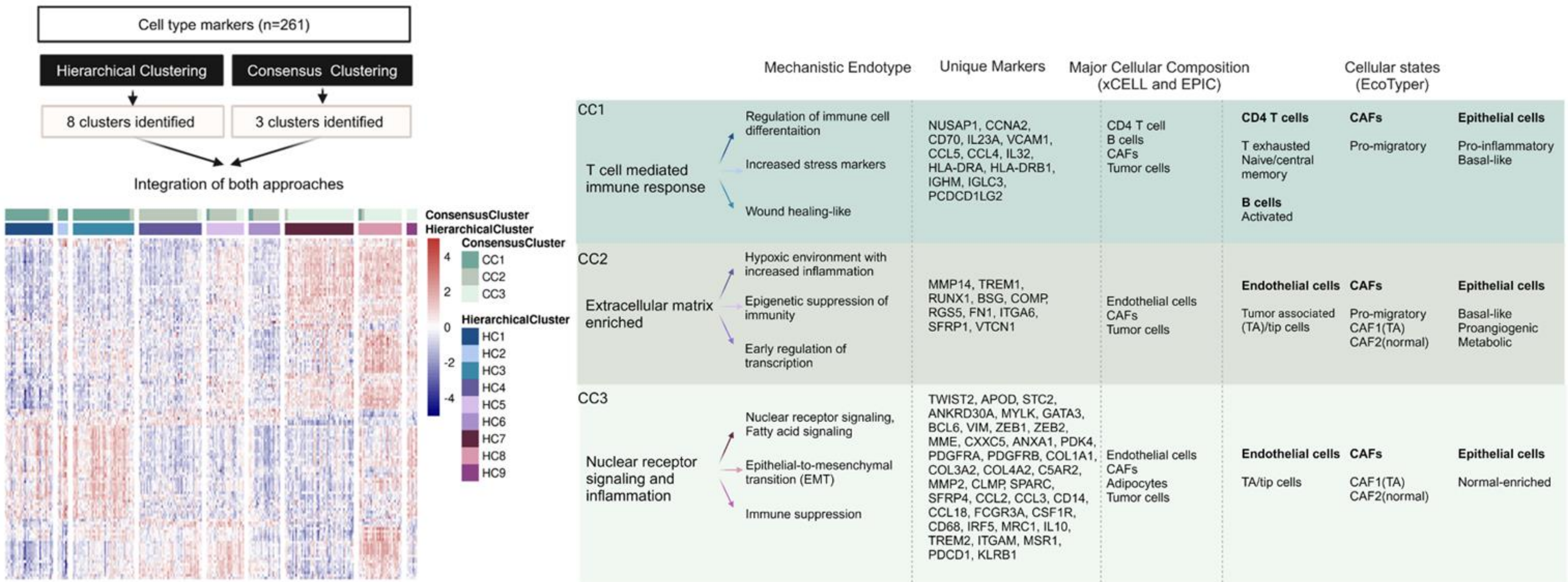
Subramaniam Lab



Downstream analysis



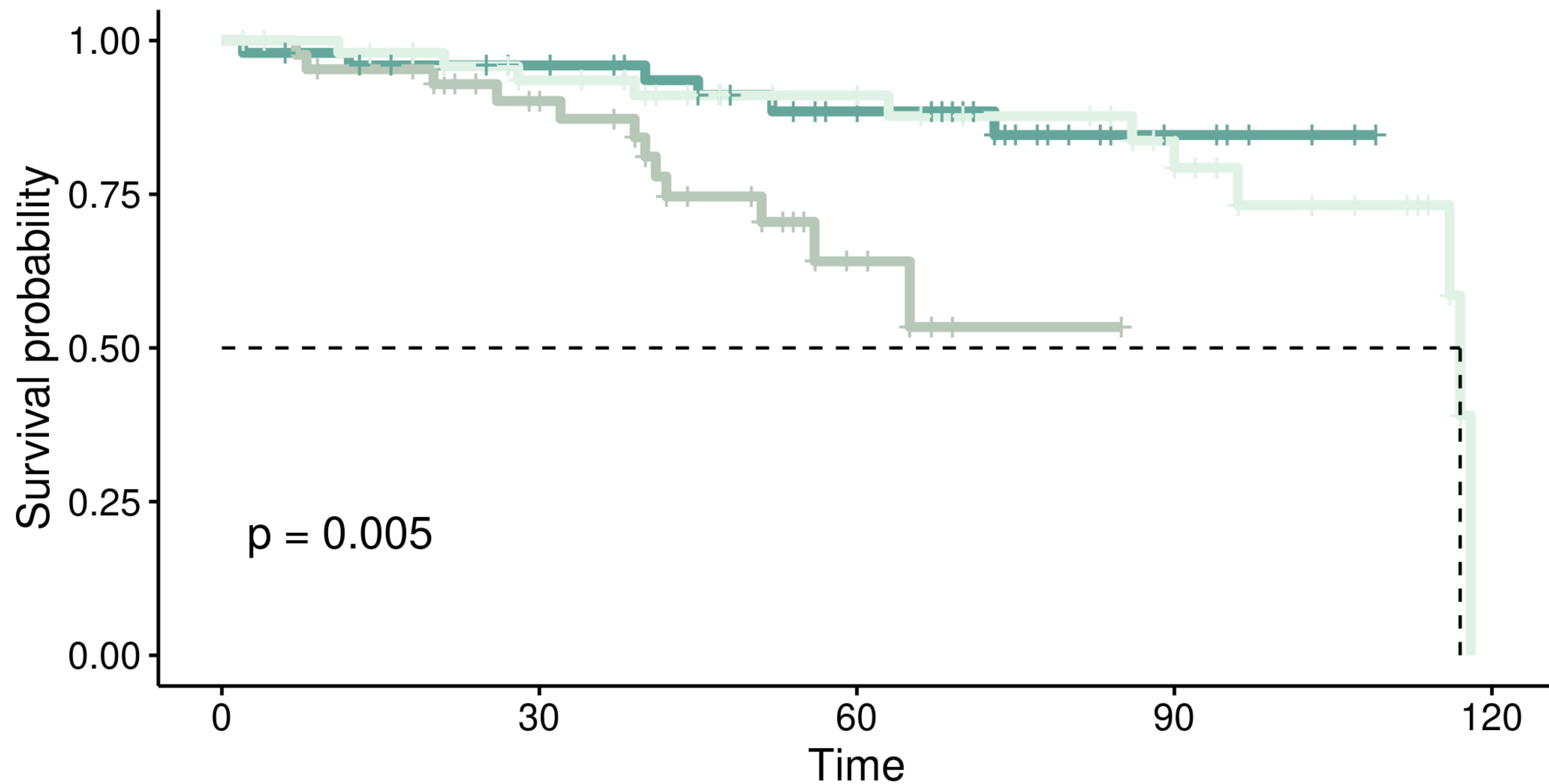
TNBC Subtypes and Endotypes based on tissue cell heterogeneity



- Hierarchical and consensus clustering divided the patients into 8 subclasses and 3 classes based on cell type marker gene expression
- Each consensus cluster contained 3 hierarchical clusters with distinct functional association and unique biomarkers

TNBC Survival Prediction and Stratification

RNASeq Profiling of Key Tx and Regulatory Factors



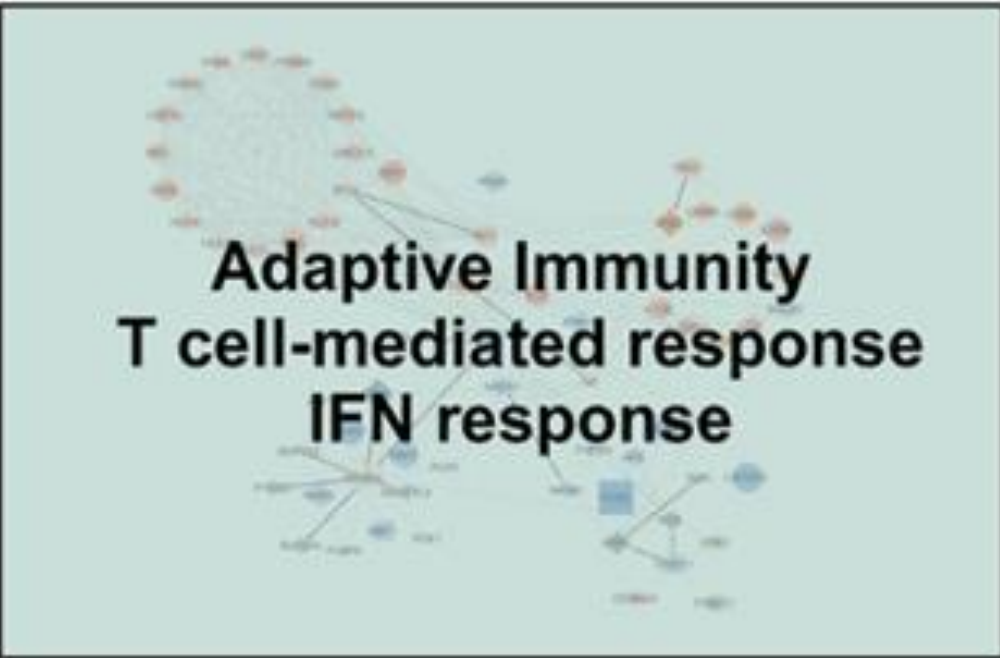
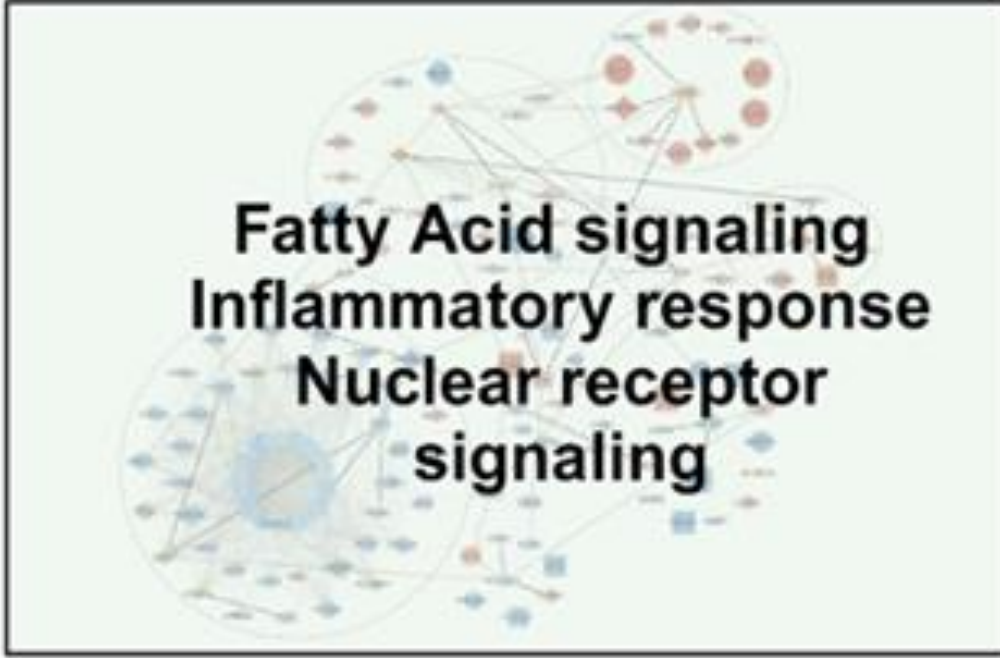
ConsensusCluster + CC1 + CC2 + CC3

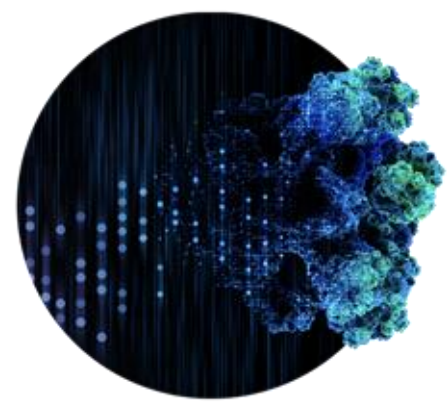
RNA Seq Hierarchical Clusters

Self Reported
Ancestry:
136 African
114 European

Proposed Trials:
CC1: Immune
Checkpoint
Blockade (ICB)
CC2: VEGFi, TGFβi
CC3: ICB + PI3Ki + NR

3 cell-type specific subtypes show distinct functional enrichments suggesting distinct treatments

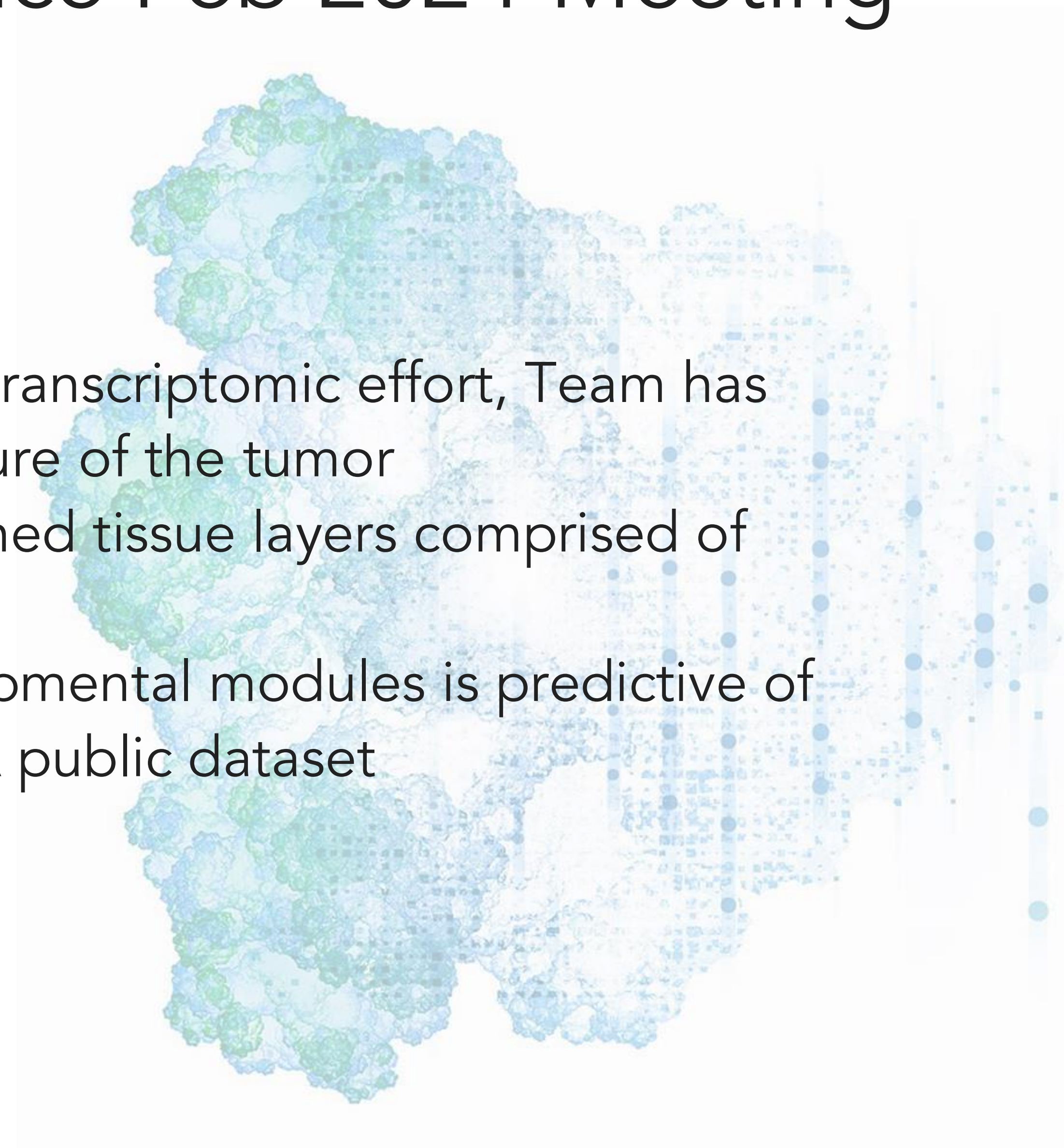
	CC1	CC2	CC3
Cell Type-based TNBC subtyping	 Adaptive Immunity T cell-mediated response IFN response	 ECM Organization Pathway	 Fatty Acid signaling Inflammatory response Nuclear receptor signaling
Literature-based immune phenotype delineation	Inflamed Phenotype Terminal exhaustion (Tex ^{term})	Excluded Phenotype	Ignored Phenotype Progenitor exhaustion (Tex ^{prog})
Potential Treatment Regimen	Anti-LAG3 Anti-TIGIT Anti-CTLA4 + ICB + Chemotherapy	Pharmacological blockade of TGFB and VEGF	ICB Inhibition of PI3Kd (Promote expansion of Tex ^{prog}) Nuclear receptor inhibitors



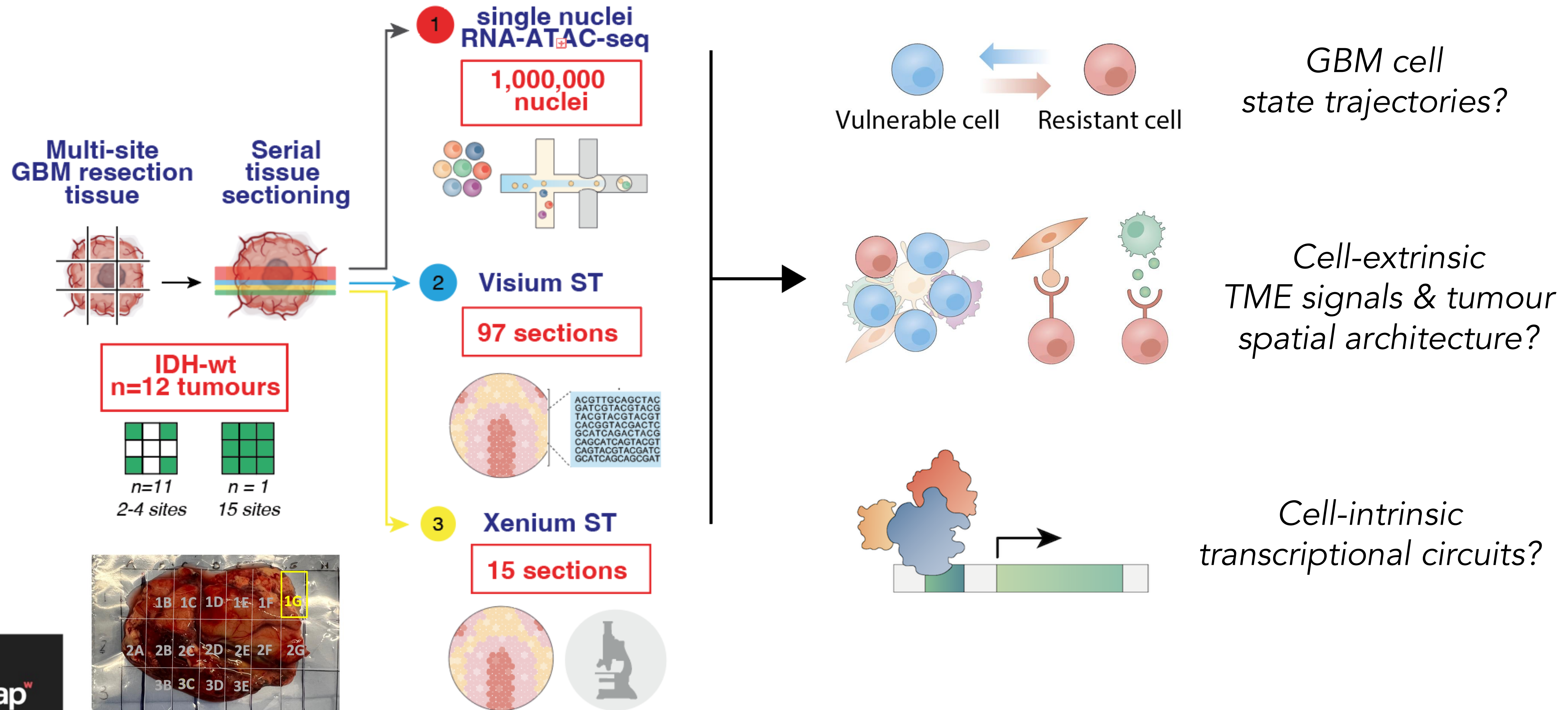
Summary of Progress since Feb 2024 Meeting

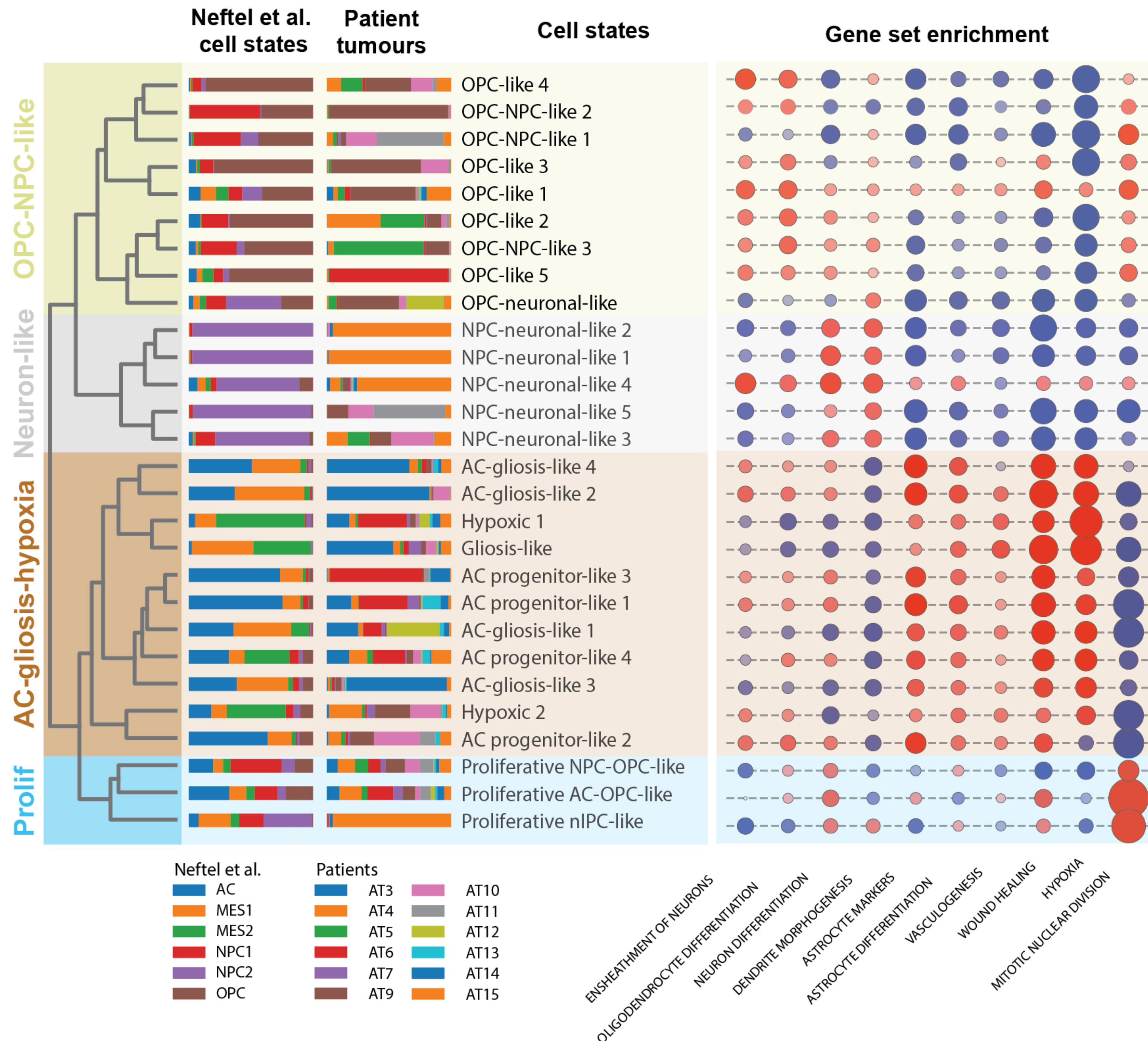
GBM- Sanger Group

- Through massive scMulti-omics and spatial transcriptomic effort, Team has defined the cellular and molecular architecture of the tumor
 - GBM tumors are organized into well-defined tissue layers comprised of specific cell states
 - Relative abundance of gliosis and developmental modules is predictive of overall survival in patients from the TCGA public dataset



GBM-space Phase 1: *deep multi-omic atlas of GBM cell states*



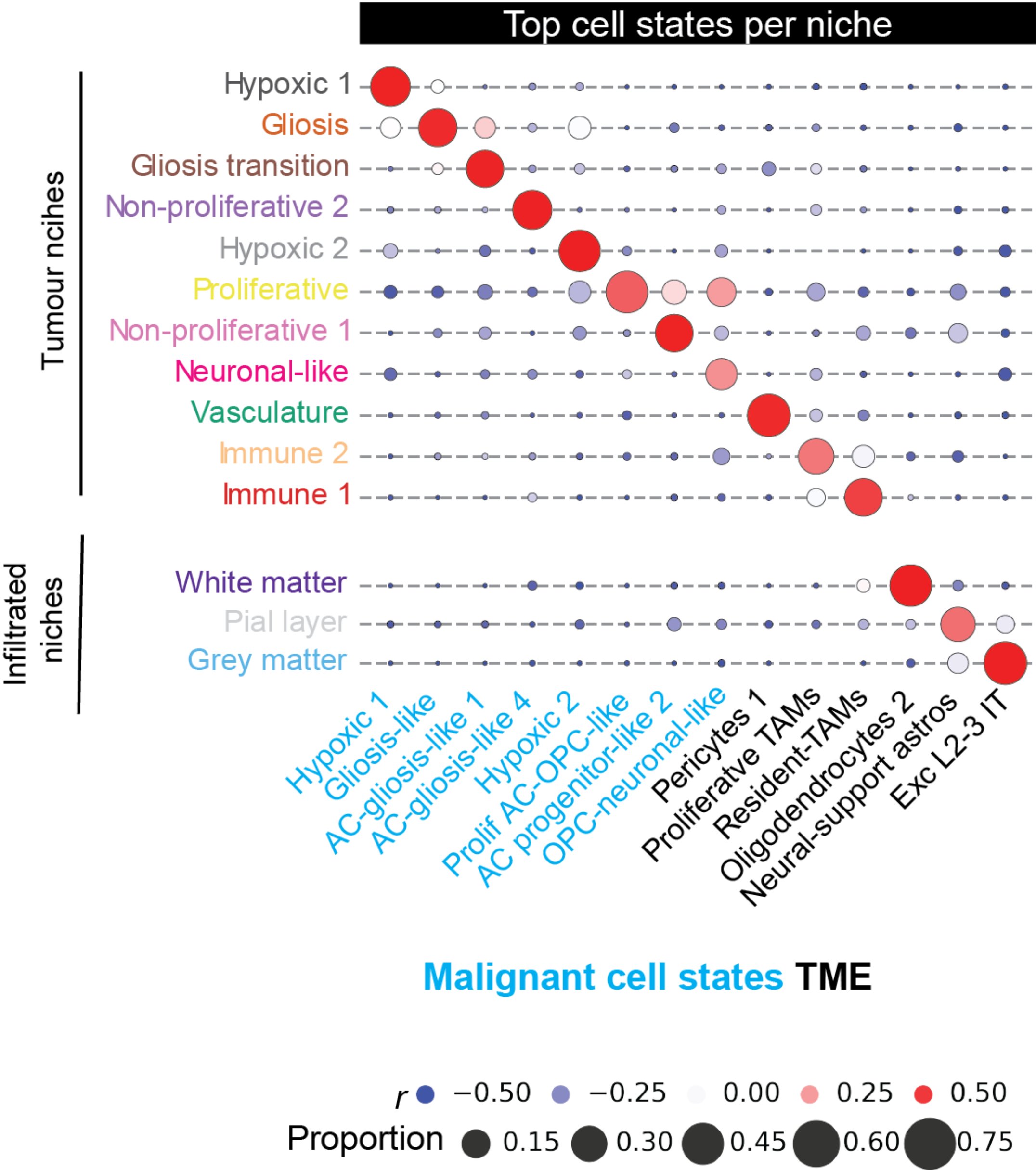
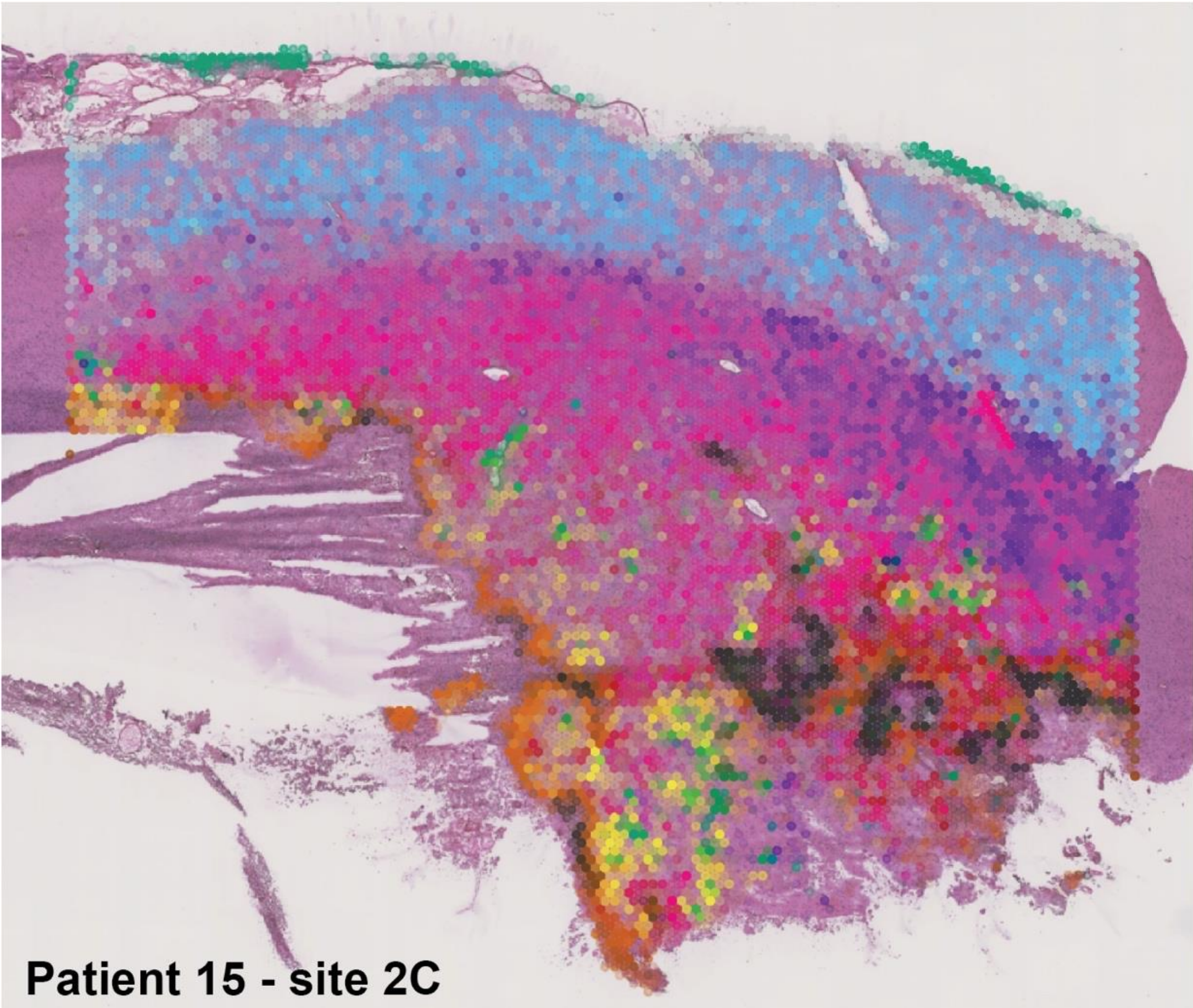


4 major groups of malignant cells spanning developmental-like to injury response and hypoxia states

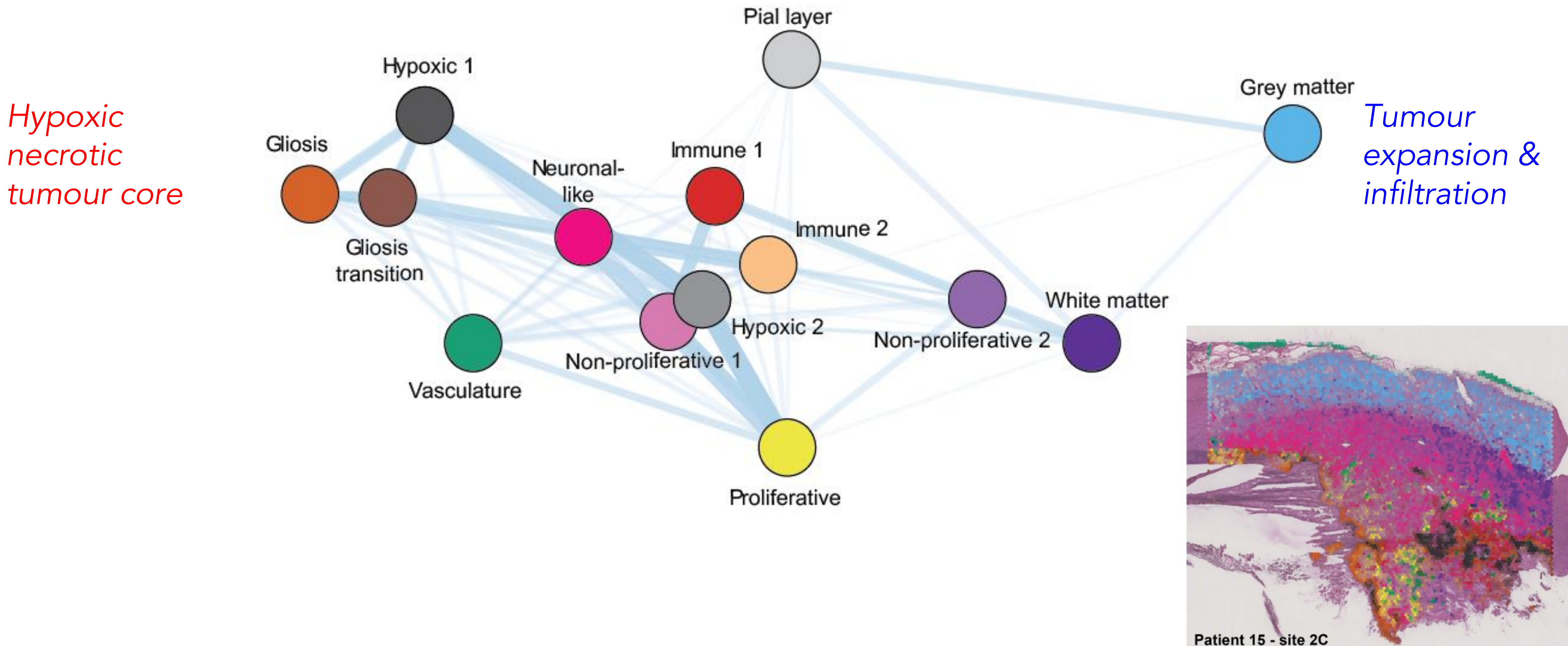
Biological annotation via relation to

- Previously described GBM states
- Enriched gene set & pathways
- Developing human brain scRNAseq

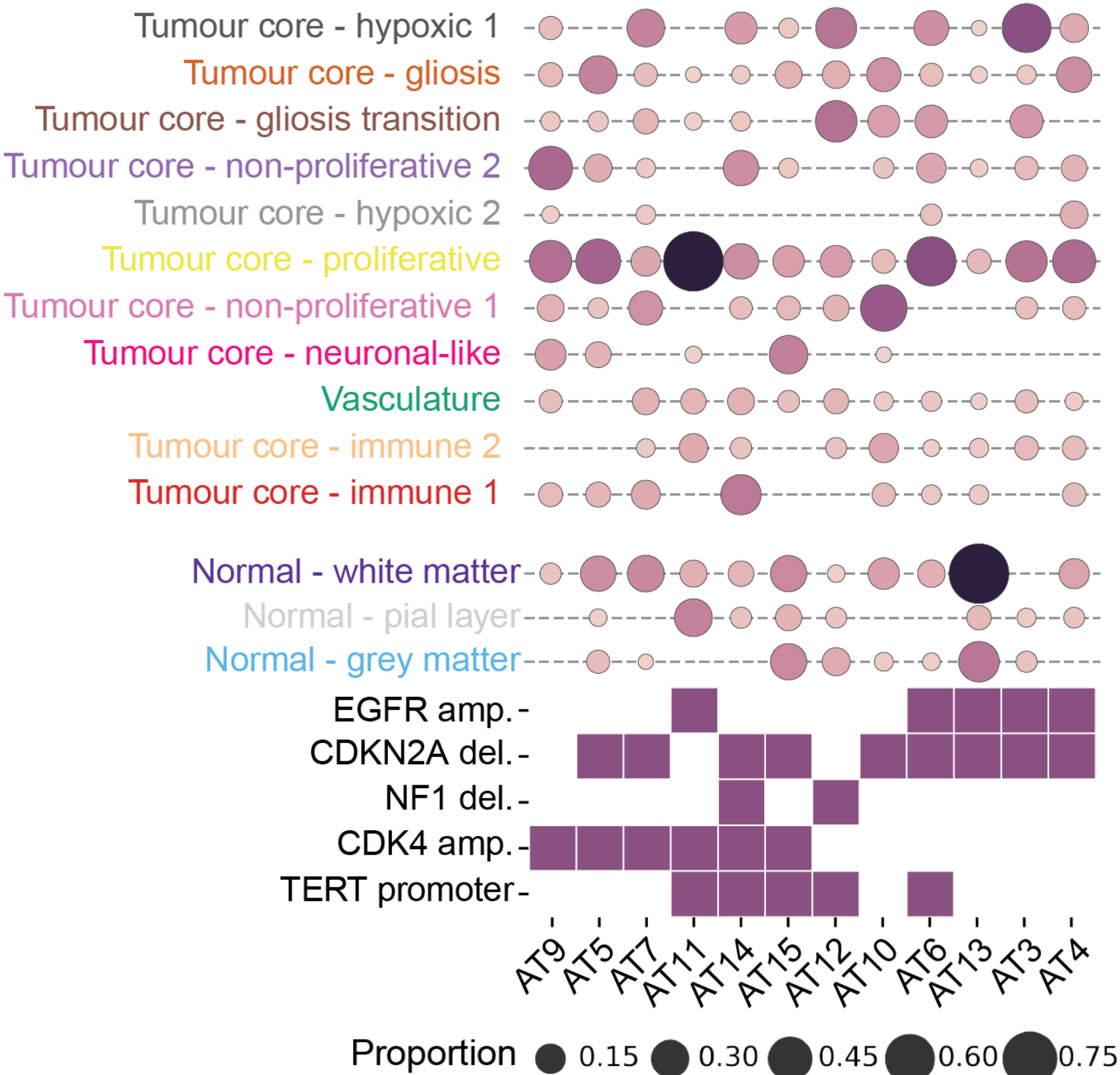
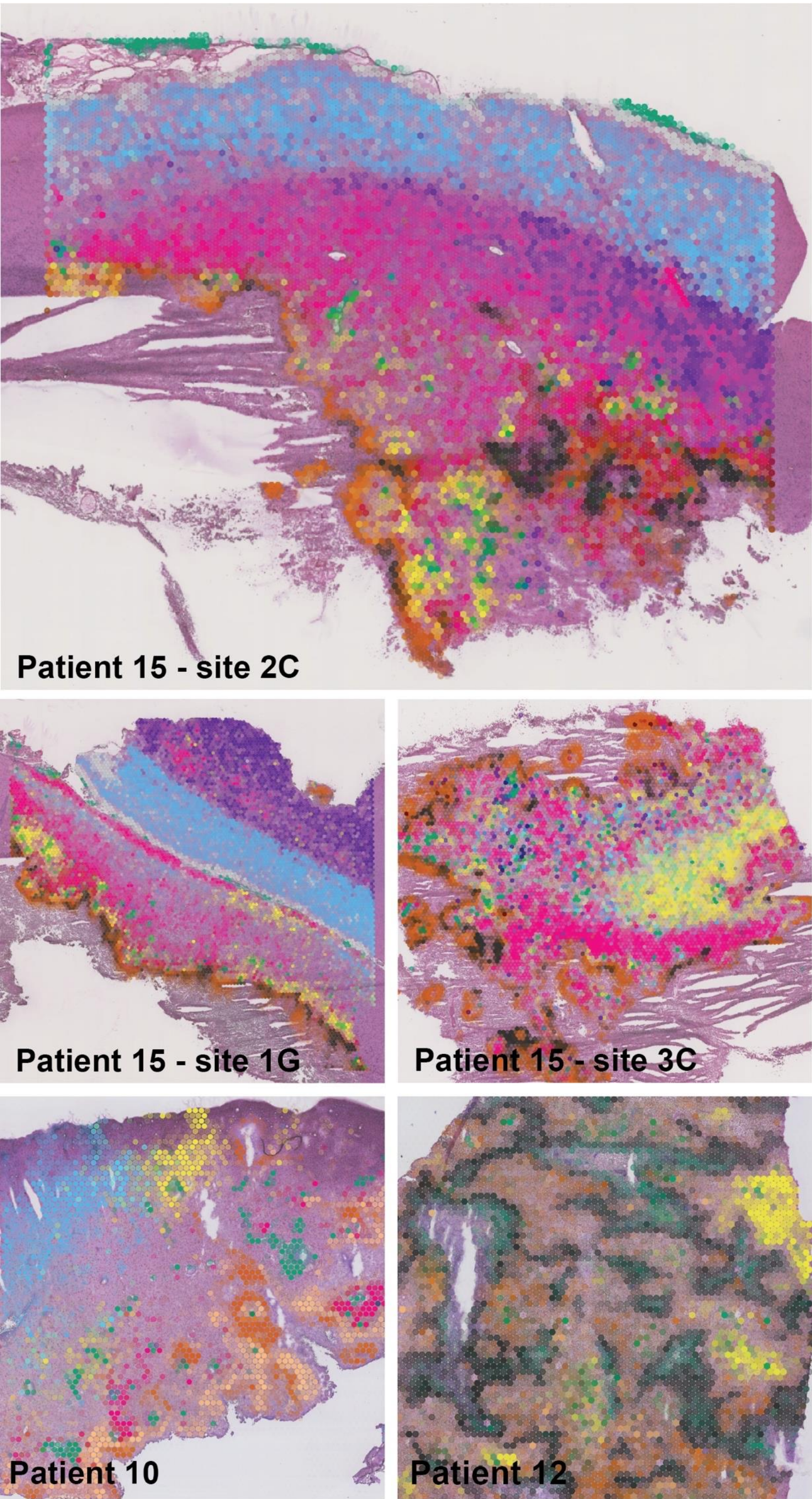
1) Most tumour niches represent spatially segregated malignant cell states



2) GBM tissue niches are organised across an anatomical axis between the invasive edge and hypoxic/gliotic core

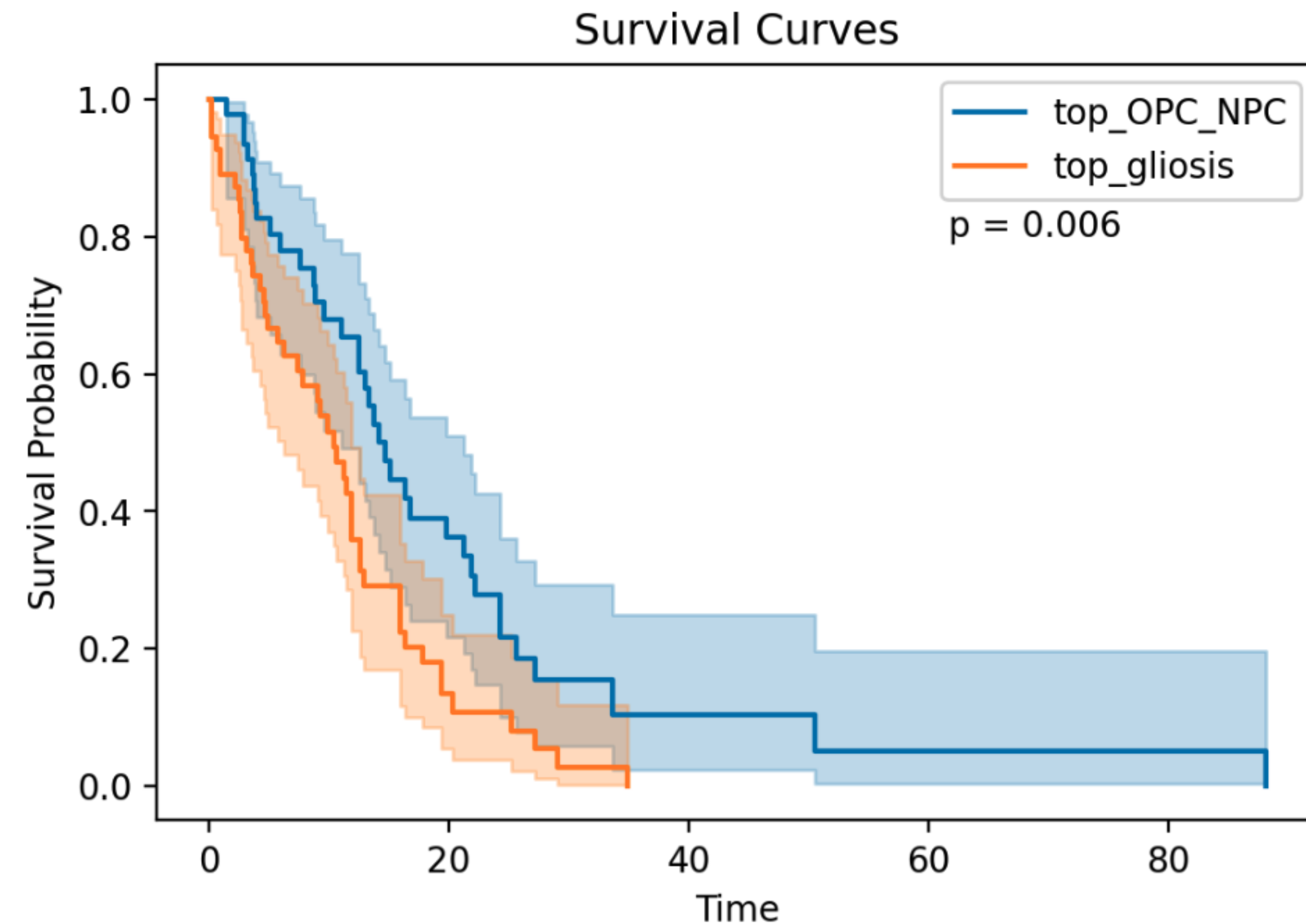


3) Many tissue niches are observed in each tumour, shared across tumour genotypes & transcriptomic subtypes



Ongoing

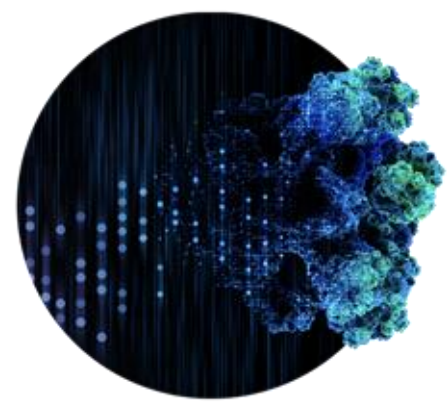
Relation of GBM malignant cell states to patient survival outcomes



Annotating **gliosis-like** VS **developmental** modules

Looking them up in TCGA and CPTAC bulk RNAseq data

IDH WT GBMs, n=100 patients

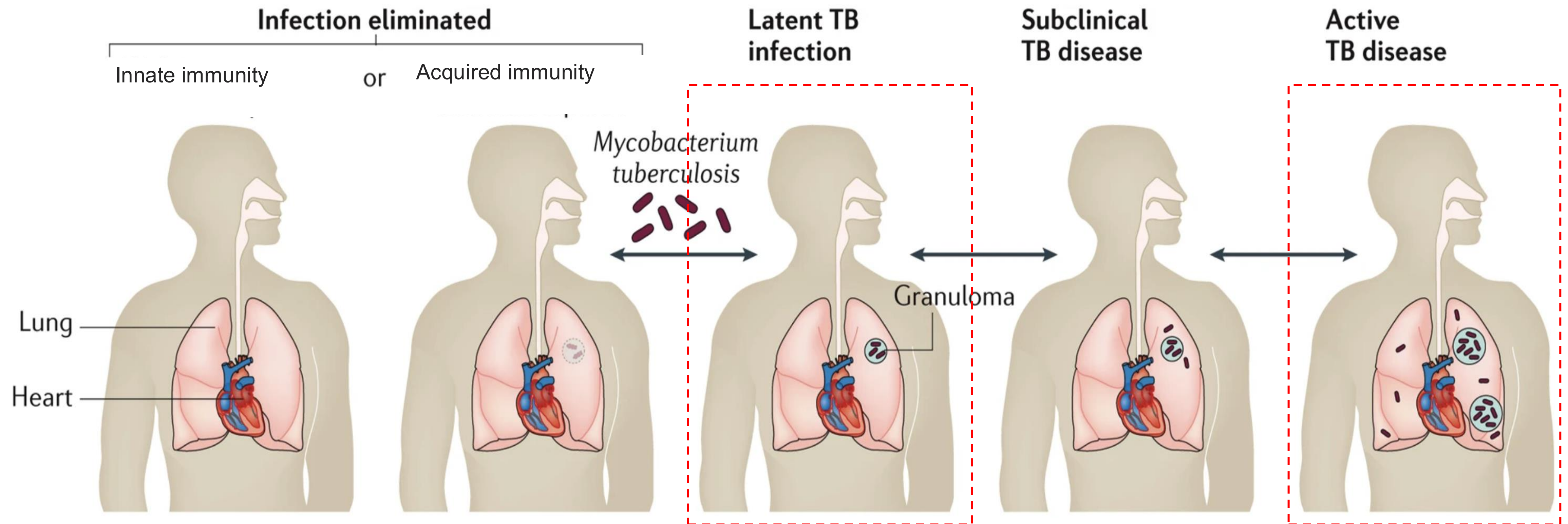


Summary of Progress since Feb 2024 Meeting

TB- UCT-Transition Project

- UCT
 - Are now collecting autopsy specimens at scale and acquiring cell and molecular data from lymph nodes
 - Have identified molecular signatures of live bacteria from people thought to be healthy/TB negative
 - Have identified bacteria in autophagosomes
 - Provided evidence for good tissue integrity in postmortem samples
 - These early results support the notion that the autopsy platform will be valuable for understanding progression from latent to active TB

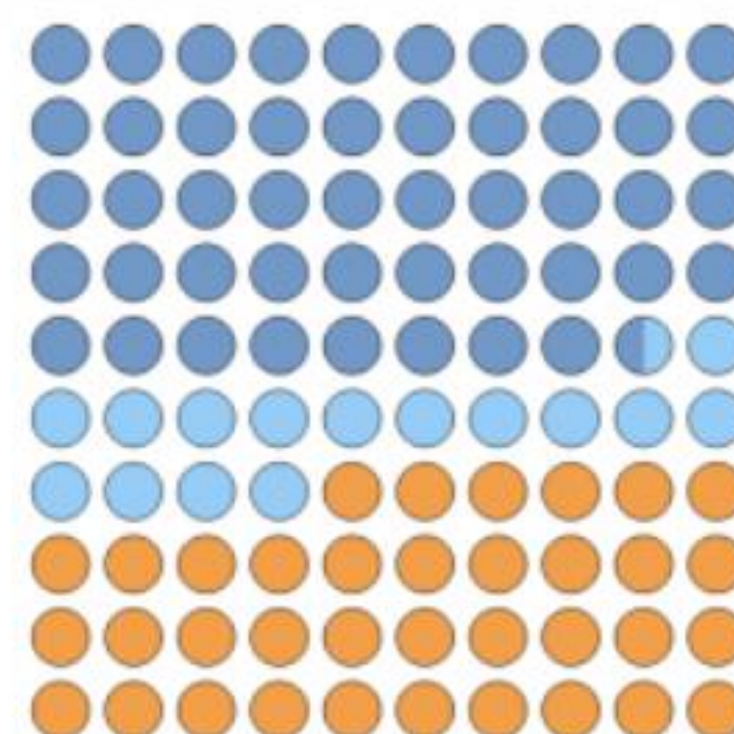
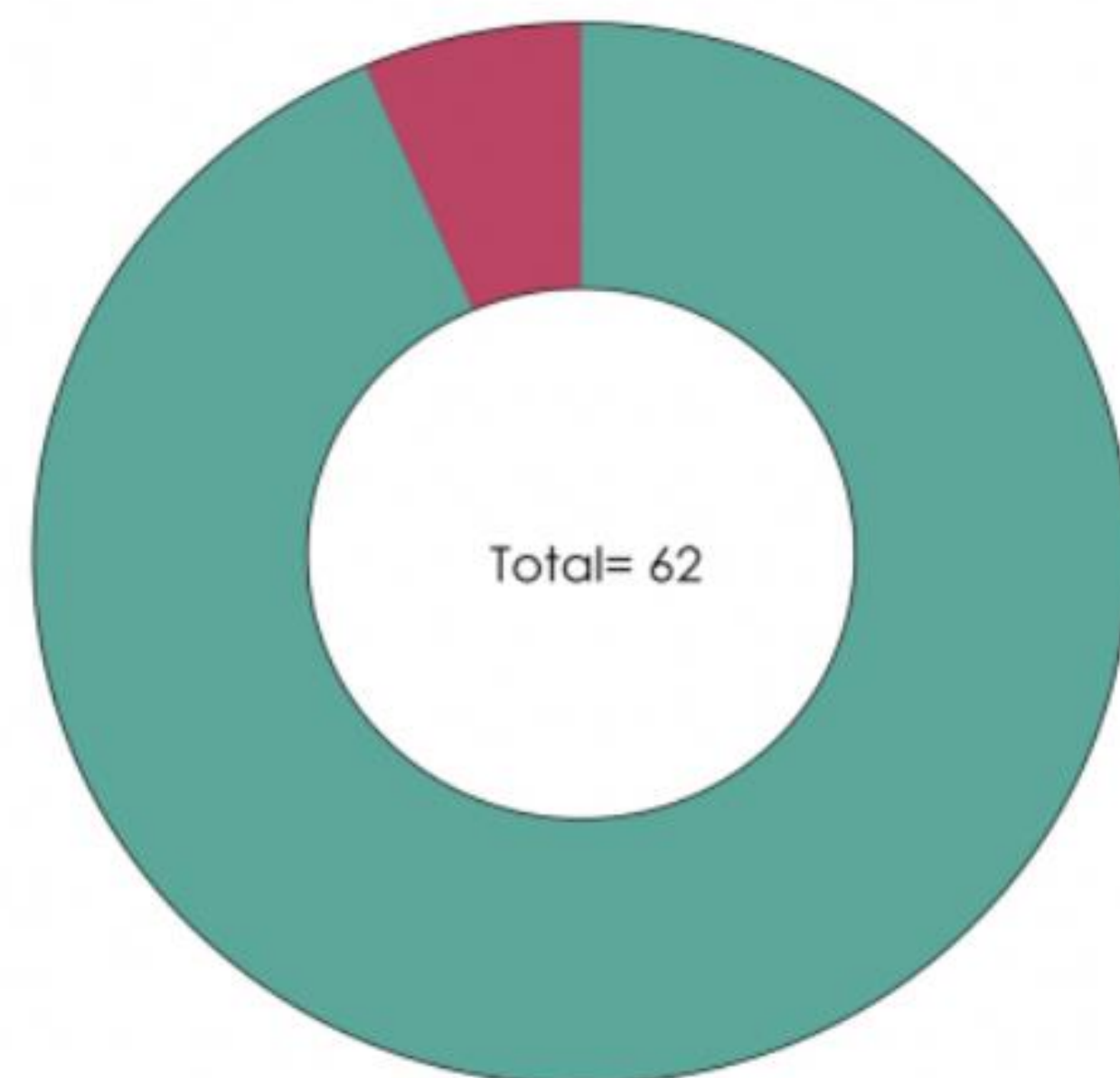
The clinical spectrum of TB infection and disease



Findings: Contextualizing molecular data to the host

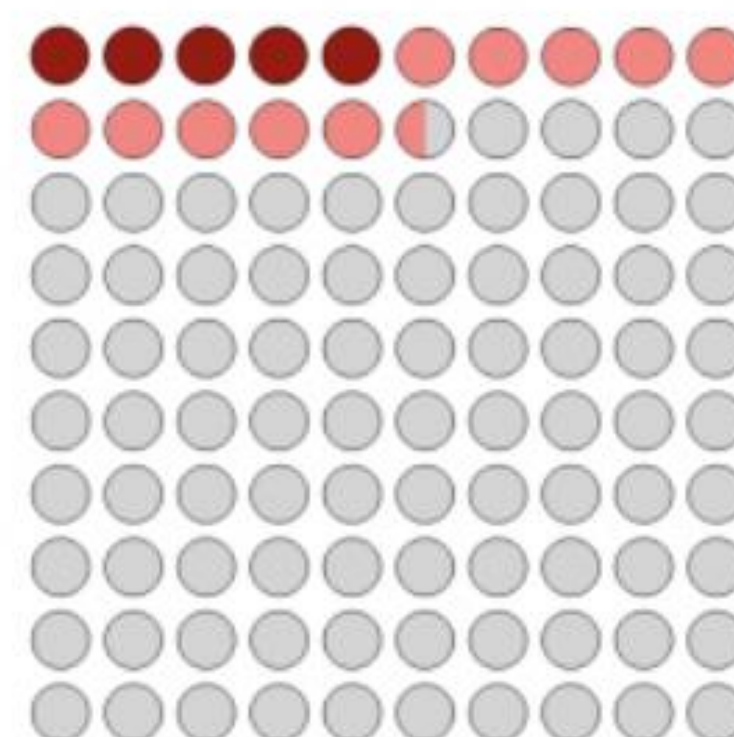
Subjects with prospective or retrospective consent from their next of kin

- 58 tested for Mtb presence
- 4 pending testing



- 48.28% Quantifiable Mtb burden in at least one LN
- 15.52% Non-quantifiable Mtb presence in any LN
- 36.21% No evidence of Mtb in either LN

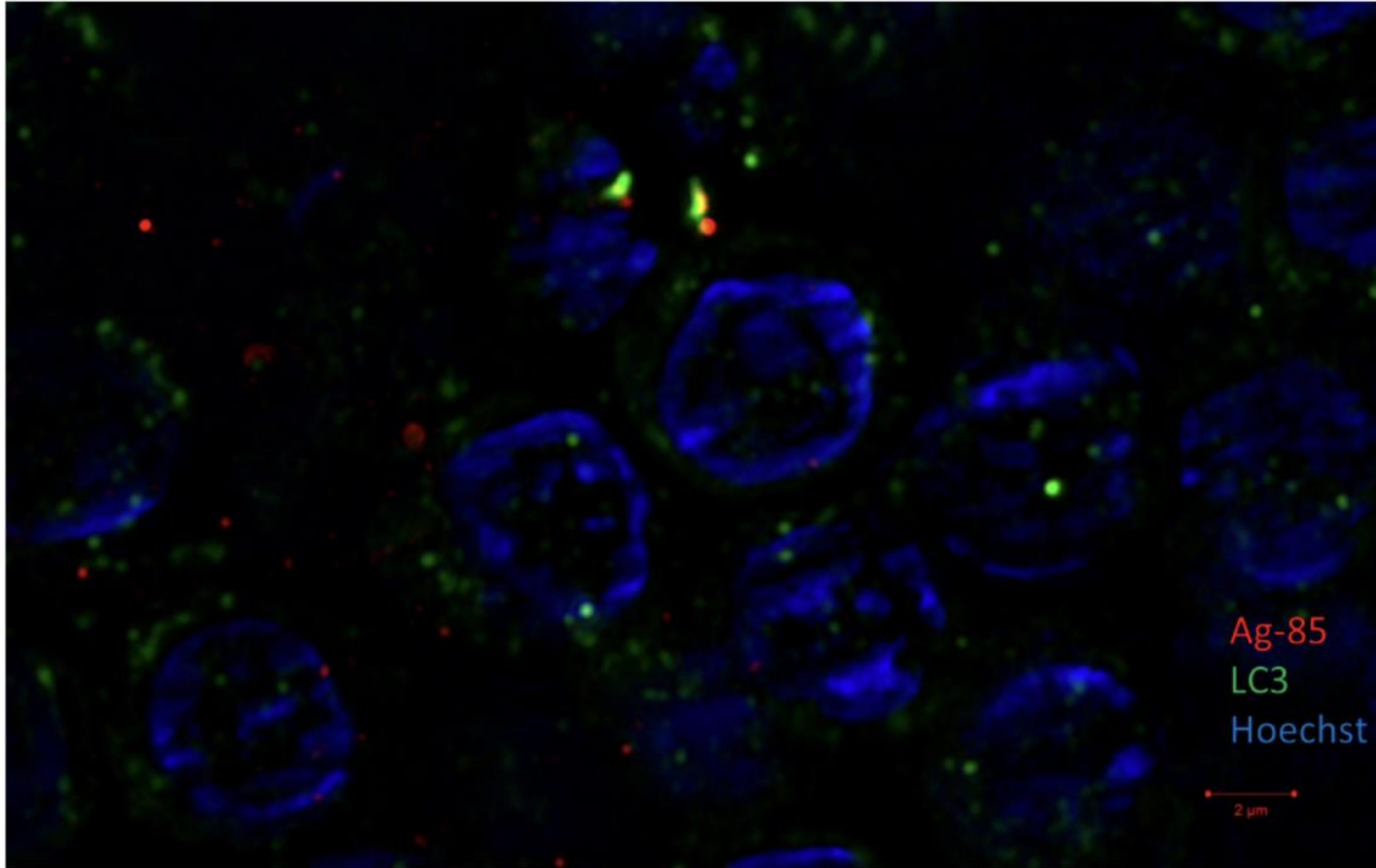
Total= 58



- 5.17% RS ratio obtained from LNs
- 15.52% Mtb RNA detected in LNs
- 84.48% No evidence of Mtb RNA in either LN

- At least one thoracic and one non-thoracic lymph node have been tested from 58 subjects
- 48.2 to 63.8% of the subjects harboured *Mtb* nucleic acid indicating their exposure to *Mtb***

Ag85-positive signal and LC3 compartment



PM016

PMI 10 hours

Clinical presentation: incidental TB

Quantiferon: +

ddPCR: DNA +

ZN: +

Ag-85
LC3
Hoechst

2 µm

LC3 phagophore recruitment - SR-SIM

PM016

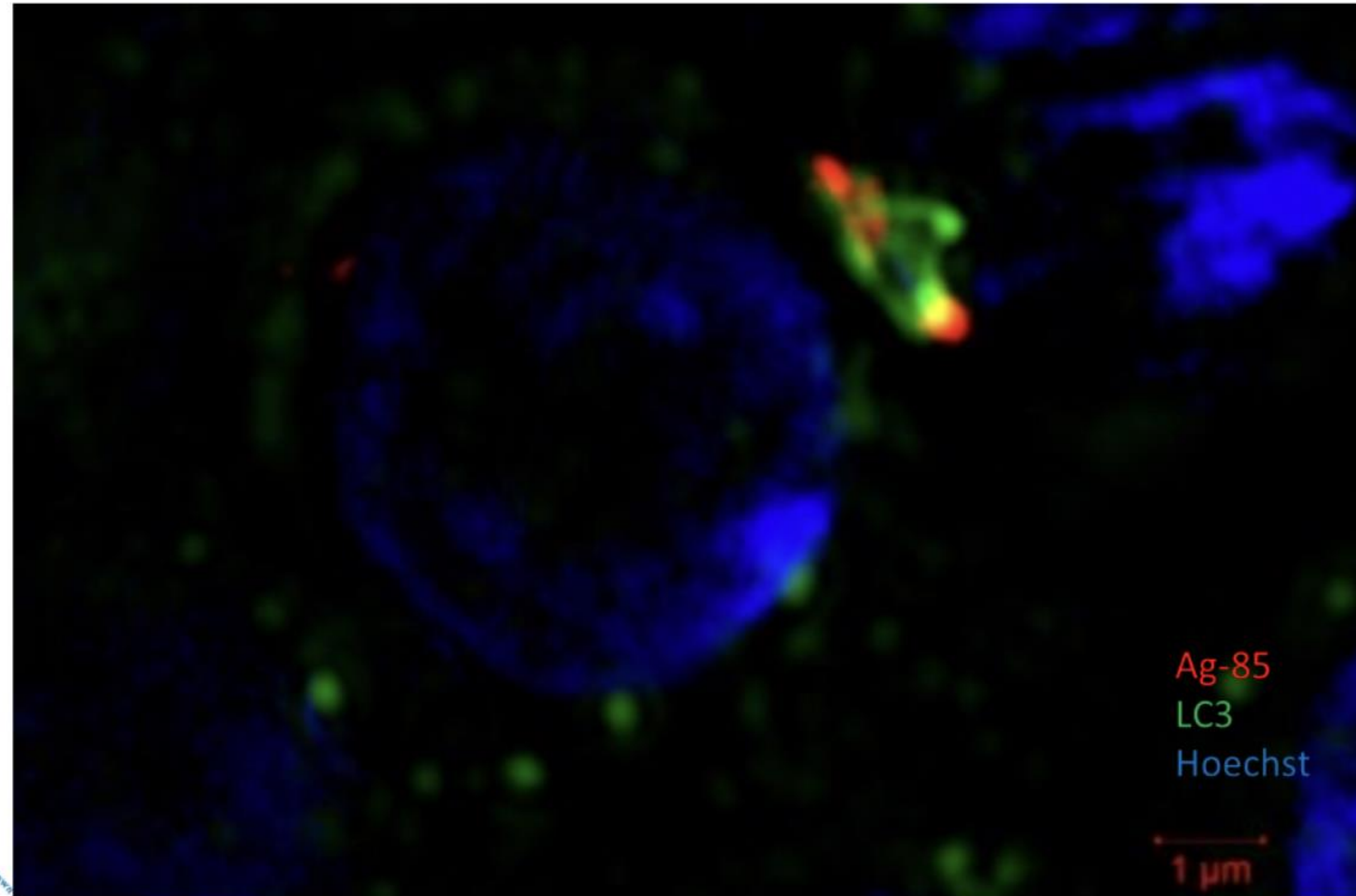
PMI 10 hours

Clinical presentation: incidental T

Quantiferon: +

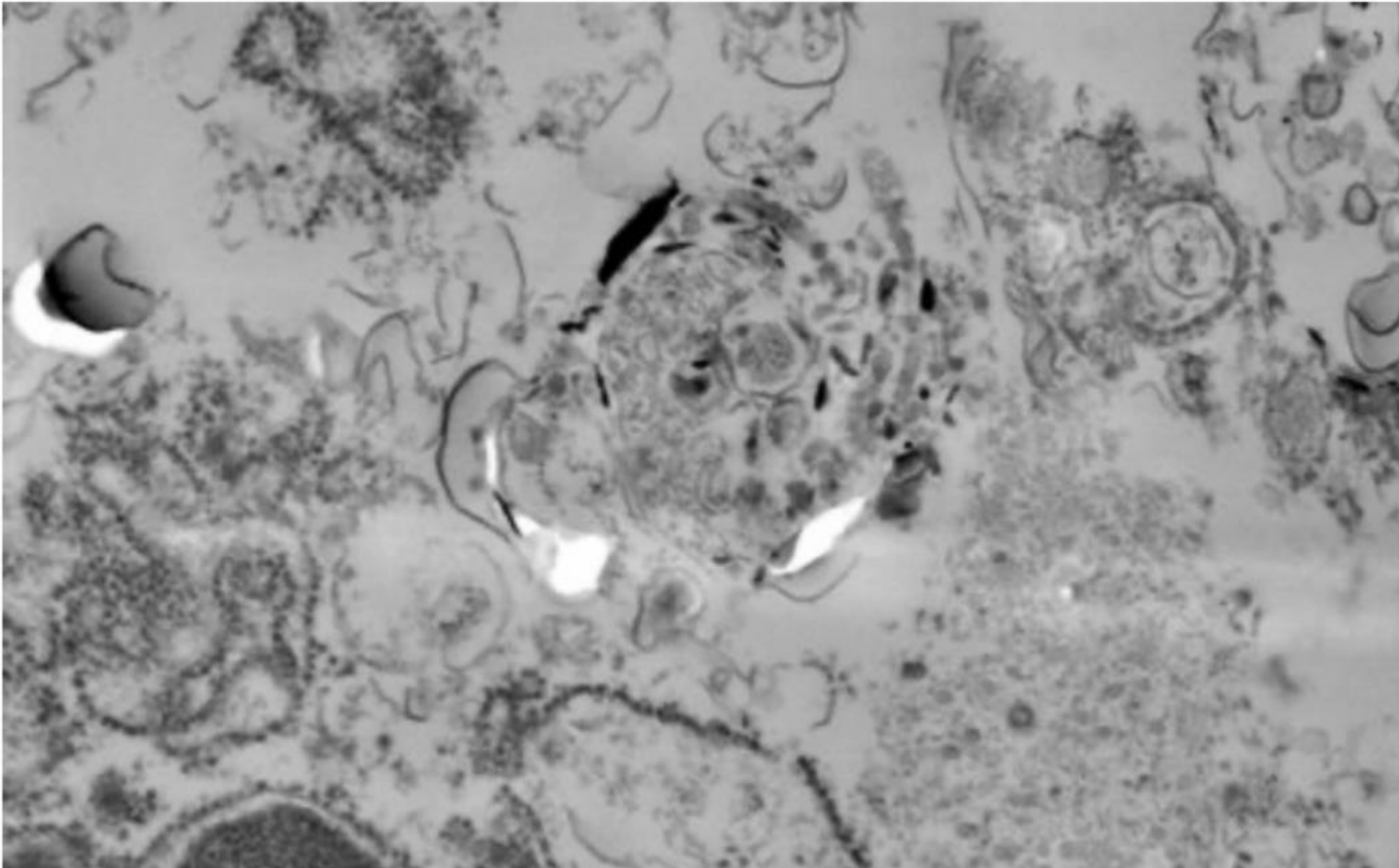
ddPCR: DNA +

ZN: +



EM-based characterisation

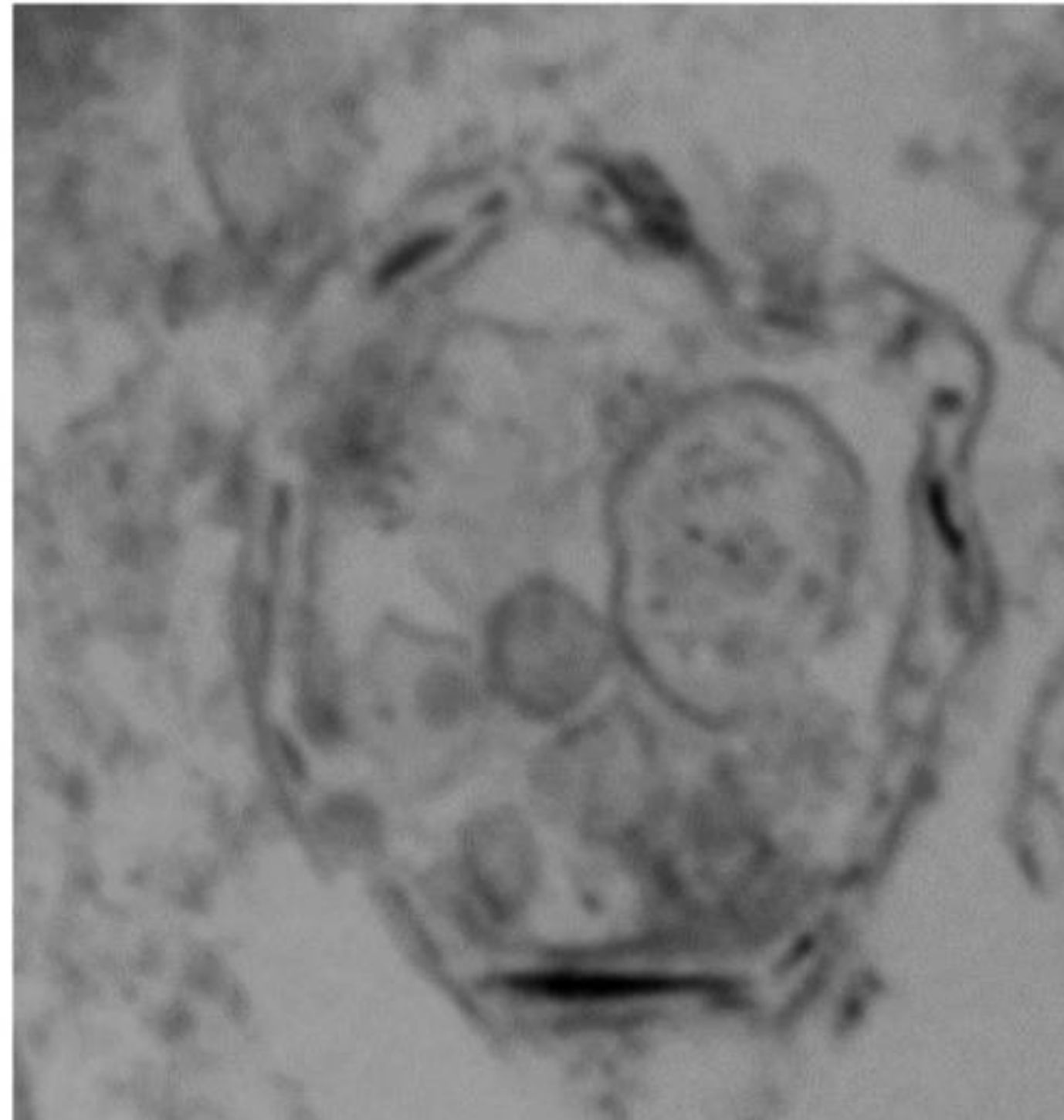
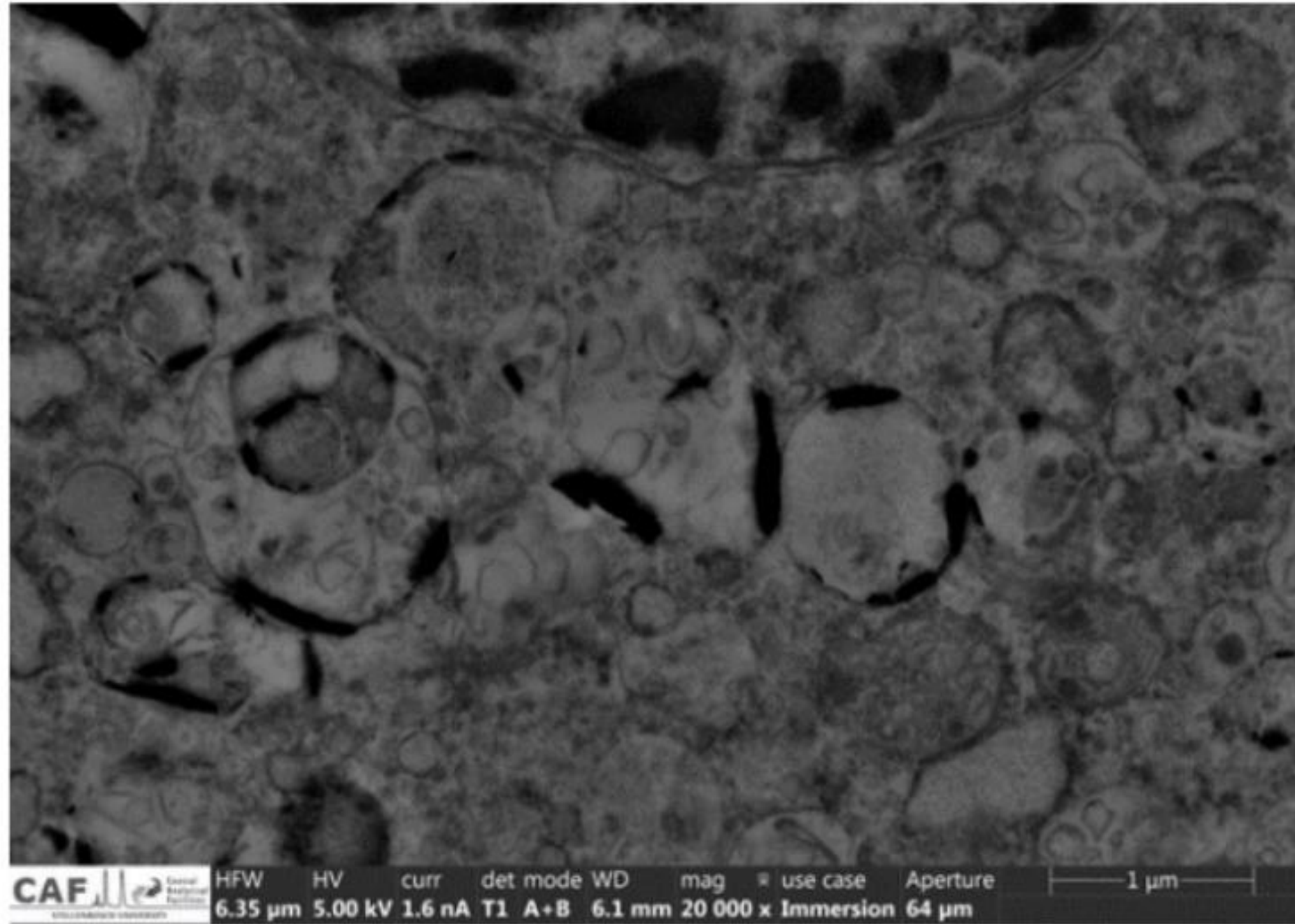
Vesicular structures, likely phagophores with presence of *Mtb* and *Mtb* fragments (high burden). Autophagy response noted



PM016
PMI 10 hours
Clinical presentation: +
Quantiferon: +
ddPCR: DNA +
ZN: +

EM-based characterisation

Membrane structures, nuclear envelope etc. are well preserved, suggesting good tissue integrity. Strong presence of vesicular structures, likely autophagic vacuoles noted.



PM016
PMI 10 hours
Clinical presentation: +
Quantiferon: +
ddPCR: DNA +
ZN: +

leap^w

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